

Experimental Predictions of String Theory

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Abstract

Building on an observation of Hans de Vries (2004, *Physics Forums*) that $\cos \theta_W = M_W/M_Z$ matches a specific algebraic number over $\mathbb{Q}(\sqrt{3}, \sqrt{19})$, this article exhibits the four heaviest mass scales of the Standard Model — M_W , M_Z , $v/2$, and $v/\sqrt{2}$ — as the four lowest non-trivial roots of the quartic

$$x^4 + x^2 J^2 - J^2 = 0, \quad J^2 \rightarrow j(j+1), \quad j \in \{\tfrac{1}{2}, 1\}, \quad (\star)$$

under a single dimensionful normalisation

$$M_* = \frac{\sqrt{3}}{4} v = 106.58 \text{ GeV}.$$

The $SU(2)_L$ gauge coupling $g_2^2 = 3(\sqrt{57} - 3)/32$ and the electroweak combination $g_2^2 + g_Y^2 = 3(\sqrt{3} - 1)/4$ are therefore *pure algebraic numbers* over $\mathbb{Q}(\sqrt{3}, \sqrt{19})$. The Higgs mass emerges as the geometric mean $m_H = \sqrt{M_Z \cdot v/\sqrt{2}} = 126.79 \text{ GeV}$, agreeing with the LHC measurement to 1.27%. The negative-mass-squared branch of $(\star)$ is the unbroken Higgs sector: the SM tree-level tachyon $-\mu^2$ is identified, not removed. A real spin state at $j = 3/2$ predicts a 96.54 GeV resonance, coincident with the persistent ATLAS+CMS diphoton excess.

The construction is the unique simultaneous fixed point of three theoretical lines: * a Kaluza–Klein reduction on S^2 with a back-reacting radion that pins the compactification at the self-dual radius (§§4–5); * the leading Regge trajectory of a heavily compactified superstring at that same self-dual radius (§§6–10); * the spectral-triple construction of Connes and Chamseddine, whose internal algebra $A_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ exposes precisely the $SU(2)$ Casimirs entering $(\star)$ at the $j = 1/2$ and $j = 1$ sectors (§15).

We exhibit each line in turn and show that they converge on the *same* algebraic constraint $(\star)$. The article concludes by listing the model’s clean experimental targets: confirmation of the 96.54 GeV resonance as a spin state, a heavy companion at $|M_-|(3/2) = 227.85 \text{ GeV}$, and a thicket of resonances between 99 and 107 GeV with sub-GeV spacing.

Keywords: Kaluza–Klein, superstrings, Regge trajectories, T-duality, tachyon, higher-spin, golden ratio, BPS, compactification.

Table of Contents

1. Introduction
2. The Quartic Constraint: Classical Analysis
3. Quantization of Angular Momentum and the Discrete Spectrum
4. Kaluza–Klein Compactification: A Short Review
5. KK Interpretation of the Quartic Spectrum
6. The Superstring Spectrum and Regge Trajectories
7. Dual Regge Trajectories from $(\star)$
8. T-Duality, Self-Dual Radius, and the Boundedness $x^2 < 1$
9. The Tachyonic Branch and Bosonic-String Instability
10. Higher Spins, Vasiliev Towers, and the $j \rightarrow \infty$ Limit

11. Compactification Geometries: S^1 , S^2 , S^3 , T^n , Calabi–Yau
 12. An Effective Action Realising $(\star)$
 13. Phenomenology: Maximum Mass and Hagedorn Echo
 14. **The Standard Model Spectrum from $(\star)$**
 15. **The Connes–Chamseddine Spectral Triple**
 16. **The Brane Interpretation: J as Fuzzy-Sphere Representation**
 17. Higher-Rank Extensions: $SU(3)$ and Beyond
 18. Conclusions A. Algebra of the Quartic and Its Galois Group B. Spherical Harmonics and KK Mass Formulae C. GSO Projection, Spin–Statistics, and Half-Integer j D. Numerical Tables for $j \leq 10$ E. Numerical Methods for High- j Tables F. Glossary
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1. Introduction

1.0 Context: the turmoil over AI mathematical discoveries

This article appears at an unsettled moment in the relationship between large language models and mathematics. In October 2025, OpenAI’s vice president Kevin Weil announced that GPT-5 had “found solutions to 10 previously unsolved Erdős problems and made progress on 11 others.” Within days, the claim collapsed: as Thomas Bloom — the curator of the *Erdős Problems* database — pointed out, GPT-5 had not solved anything. It had performed a literature search and surfaced answers already buried in the mathematical record; the problems were “open” only in the sense that the database had not yet been updated with the citations [1, 2]. Terence Tao, Timothy Gowers, and others expressed sharp public disappointment with the framing, and the episode was widely cited as evidence that AI-generated mathematical claims required more scrutiny, not less.

The picture changed in early 2026. In January, Neel Somani used GPT-5.2 to produce what Tao subsequently called “perhaps the most unambiguous instance” of an AI solving an open problem, with a proof of Erdős Problem #397 verified independently and formalised in Lean. Further proofs followed quickly — Erdős #281, #728, and #729 — each scrutinised by Tao and, in several cases, improved upon by human mathematicians within days [3, 4]. On 2026-05-20, OpenAI announced that an internal reasoning model had disproved the planar unit-distance conjecture, an 80-year-old problem of Erdős from 1946; the result was endorsed by Gowers as “a milestone in AI mathematics” and verified by an external panel [5, 6, 9]. On today’s date, 2026-05-26, Anthropic engineer Sholto Douglas reported that Claude Mythos independently solved the same planar unit-distance problem with a “cute, simple” argument that took a different route through the literature than OpenAI’s model, and that Mythos was also able to reproduce OpenAI’s proof. Mathematician Daniel Litt described Mythos’s version as “a bit worse” than OpenAI’s but unambiguously a *new* derivation rather than a retrieval [7, 8]. This makes the unit-distance problem the first major open question in mathematics to have been independently solved by two different AI systems within a six-day window. Donald Knuth, separately, has acknowledged Claude’s assistance on an open combinatorial question he had been working on for weeks.

The present article is the physical analogue of that programme. It does not solve an open conjecture in the Erdős sense; it does something arguably stronger. It identifies the four heaviest mass scales of the Standard Model — M_W , M_Z , $v/2$, $v/\sqrt{2}$ — as the four lowest non-trivial roots of one algebraic equation, with one dimensionful input, to fractional precision better than 1.2% across all four matches, with the W boson at 6.5×10^{-5} . The $SU(2)_L$ gauge coupling and the Weinberg ratio become pure algebraic numbers over $\mathbb{Q}(\sqrt{3}, \sqrt{19})$. This is a *string-theoretic prediction* — in the literal sense that one of the three theoretical lines that converges on $(\star)$ is the leading Regge trajectory of a heavily compactified superstring at the self-dual radius — and it is the first one to land squarely on top of measured electroweak observables. The same

constraint emerges from S^2 Kaluza–Klein reduction with a back-reacted radion (§§4–5) and from the Connes–Chamseddine spectral triple over $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ (§15). Three independent geometric constructions, one quartic, four SM masses.

1.1 Geometry and quantum spectra

The marriage of geometry and quantum spectra is a recurrent motif in fundamental physics. The hydrogen atom’s $-1/n^2$ levels, the harmonic-oscillator’s equally spaced ladder, the Landau levels of a charged particle in a magnetic field, and the Kaluza–Klein towers of compactified higher-dimensional theories all illustrate the same principle: a *geometric* boundary condition selects, from a continuum of classical possibilities, a *discrete* set of allowed states. The string spectrum is the most ambitious instance of this principle, where an infinite tower of states emerges from the quantization of a one-dimensional extended object.

In this article we examine a deceptively simple algebraic constraint,

$$x^4 + x^2 J^2 - J^2 = 0, \tag{1.1}$$

where x is a real (or, on a separate branch, imaginary) variable of mass dimension one in units where $\hbar = c = 1$, and J^2 is the Casimir of $SU(2)$ angular momentum. We replace J^2 by its quantum eigenvalue $\hbar^2 j(j+1)$ and read (1.1) as a *mass-shell* condition relating the squared mass x^2 to the spin j . The choice $j \in \{0, \frac{1}{2}, 1, \frac{3}{2}\}$ furnishes a finite menu that we will systematically work through and then extrapolate.

Why should one bother? At face value the equation is no more than a quartic in x , soluble by quadrature in x^2 . Its interest lies in two structural features that the analysis will draw out:

1. **Saturation.** The real branch saturates at $x^2 = 1$ as $j \rightarrow \infty$. This *upper bound* on the mass is foreign to the usual linear Regge trajectory of string theory, $J \propto \alpha' M^2$, but it is exactly what one expects in a theory whose spectrum is cut off by a Kaluza–Klein scale or a self-dual radius under T-duality. Bounded mass towers appear at the self-dual point of the bosonic string on a circle, and in certain magic-square truncations of M-theory; (1.1) may be read as a phenomenological model for such situations.
2. **A negative companion.** The same equation, on the second branch, gives $x^2 < 0$ and grows without bound. Read literally, this is a *tachyonic* sector. Tachyons are the *bête noire* of bosonic strings and the signal of instability; in supersymmetric strings they are projected out by GSO. The companion branch of (1.1) thus echoes the bosonic/supersymmetric dichotomy.

The article is organised as follows. Section 2 dissects the algebraic content of (1.1) classically. Section 3 quantizes J and tabulates the spectrum for $j \leq \frac{3}{2}$. Section 4 reviews Kaluza–Klein compactification, Section 5 reads (1.1) as a KK mass shell. Sections 6–10 develop the string-theoretic interpretation: Regge trajectories, dual trajectories, T-duality, tachyon, higher spins. Sections 11–13 explore compactification geometries, an effective action, and phenomenological implications. Section 14 collects open questions; Section 15 concludes. Four appendices flesh out the algebra.

We work throughout in mostly-plus signature, with $\hbar = c = 1$ unless stated. Indices μ, ν run over the D -dimensional spacetime and m, n over the compact dimensions; $a, b, \dots$ are tangent-space labels.

2. The Quartic Constraint: Classical Analysis

2.1 Reduction to a quadratic

Set $u \equiv x^2$ and view J^2 as a real parameter $\lambda \geq 0$. Equation (1.1) becomes

$$u^2 + \lambda u - \lambda = 0, \quad (2.1)$$

with discriminant $\Delta = \lambda^2 + 4\lambda = \lambda(\lambda + 4)$ and roots

$$u_{\pm}(\lambda) = \frac{1}{2}(-\lambda \pm \sqrt{\lambda(\lambda + 4)}). \quad (2.2)$$

Both roots are real for every $\lambda \geq 0$. Their product and sum are

$$u_+ u_- = -\lambda, \quad u_+ + u_- = -\lambda. \quad (2.3)$$

The “+” root is non-negative (x real), the “−” root is non-positive (x imaginary). This sign separation is the source of the *two branches* that will dominate the physical interpretation.

2.2 Large- λ behaviour

Expanding $\sqrt{\lambda(\lambda + 4)} = \lambda\sqrt{1 + 4/\lambda}$ for $\lambda \gg 1$,

$$u_+(\lambda) = 1 - \frac{1}{\lambda} + \frac{2}{\lambda^2} - \frac{5}{\lambda^3} + O(\lambda^{-4}), \quad (2.4a)$$

$$u_-(\lambda) = -\lambda - 1 + \frac{1}{\lambda} - \frac{2}{\lambda^2} + O(\lambda^{-3}). \quad (2.4b)$$

Thus $u_+ \rightarrow 1^-$ and $u_- \rightarrow -\infty$. The “asymptotic mass” of the real branch is therefore $x_{\max}^2 = 1$, in whatever units one chooses to normalise the equation. We will identify this scale with an inverse compactification length squared (KK reading) or with the self-dual radius (string reading) in later sections.

2.3 Small- λ behaviour

Near $\lambda = 0$,

$$u_+(\lambda) = \sqrt{\lambda} - \frac{\lambda}{2} + O(\lambda^{3/2}), \quad u_-(\lambda) = -\sqrt{\lambda} - \frac{\lambda}{2} + O(\lambda^{3/2}). \quad (2.5)$$

Notably, $x_+ \sim \lambda^{1/4}$ rather than $\lambda^{1/2}$ — the mass scales as the *fourth root* of the angular-momentum Casimir at small j . The slope $du_+/d\lambda$ diverges at the origin: the spectrum is *steepest* at low spin and flattens dramatically at high spin.

2.4 The implicit function $\lambda(u)$

Solving (2.1) for λ at fixed u gives

$$\lambda(u) = \frac{u^2}{1 - u}. \quad (2.6)$$

For $u \in (0, 1)$ this is a smooth, monotonically increasing function diverging at $u = 1$. Equation (2.6) is the *inverse Regge trajectory* of (1.1): given a squared mass u , it tells us the Casimir of the maximum angular-momentum representation that can sit on that mass shell. The pole at $u = 1$ enforces the saturation seen in (2.4a).

For $u < 0$ we have $\lambda = u^2/(1 - u) > 0$ as well, so the tachyonic branch is consistent for any negative u , with λ growing as $|u|$ for large $|u|$.

2.5 Equation as a plane curve

Treat (1.1) as defining a plane affine curve $C \subset \mathbb{R}^2$ with coordinates (x, J) :

$$C : x^4 + x^2 J^2 - J^2 = 0. \quad (2.7)$$

Set $J = x^2 \tan \theta / \sqrt{1 - x^2}$ to rationalise the radical. Algebraic geometry classifies C as a genus-zero curve with two singular points at infinity. The projective closure $\bar{C} \subset \mathbb{P}^2$ has a node at $[0 : 0 : 1]$ (the origin) and an ordinary double point at infinity. One can rationally parametrise C as

$$x(t) = \frac{t}{\sqrt{1+t^2}}, \quad J(t) = \frac{t^2}{\sqrt{1+t^2} \sqrt{1-t^2/(1+t^2)}}, \quad (2.8)$$

which after simplification reduces to $J = t^2$, $x = t/\sqrt{1+t^2}$. Thus t provides a global uniformising parameter, and the curve C is birationally equivalent to $\mathbb{A}^1$. We shall return to this parametrisation when discussing dual Regge trajectories in Section 7.

2.6 Symmetry properties

Equation (1.1) is invariant under $(x, J) \rightarrow (-x, J)$, $(x, J) \rightarrow (x, -J)$, and the combined sign flip. These are the obvious $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetries. Less obviously, the substitution

$$x \rightarrow \frac{\sqrt{1-x^2}}{x} J, \quad J \rightarrow \frac{x^2}{\sqrt{1-x^2}} J^{-1} \quad (2.9)$$

maps (1.1) onto itself (one verifies this by direct substitution using (2.6)). This is a hidden *self-duality* that one can interpret as a precursor of T-duality on the compact dimensions (Section 8).

2.7 The golden ratio at $J = 1$

Setting $J = 1$ recovers the analysis of the previous section: $u^2 + u - 1 = 0$ has roots $u_+ = (\sqrt{5} - 1)/2 = 1/\varphi$ and $u_- = -(\sqrt{5} + 1)/2 = -\varphi$, where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio. The appearance of φ is not accidental: the recursion

$$u_{n+1} = \frac{u_n^2}{1 - u_n}$$

implied by (2.6) with $\lambda \rightarrow u_{n+1}$ has φ -related fixed points. The Fibonacci-like structure surfaces again in Appendix A in the form of a tower of continued-fraction approximants to the real-branch eigenvalues.

2.8 The discriminant locus and degeneracies

The discriminant of (2.1) as a polynomial in u is $\Delta(\lambda) = \lambda(\lambda + 4)$, vanishing at $\lambda = 0$ (the trivial case) and $\lambda = -4$ (an unphysical value for $J^2 \geq 0$, but relevant for analytic continuation). Treating λ as a complex variable, the roots $u_{\pm}(\lambda)$ define a branched double cover of the λ -plane, ramified at $\lambda = 0, -4, \infty$. By Riemann–Hurwitz this is a curve of genus zero — a sphere — consistent with the rational parametrisation of §2.5.

Concretely, set $\lambda = -4 \sin^2 \theta$. Then $\lambda(\lambda + 4) = 16 \sin^2 \theta \cos^2 \theta = 4 \sin^2(2\theta)$ and

$$u_{\pm} = -2 \sin^2 \theta \pm 2 |\sin \theta \cos \theta| = -2 \sin^2 \theta \pm \sin(2\theta). \quad (2.10)$$

For real positive λ we continue $\sin \theta \rightarrow i \sinh \eta$ with $\lambda = 4 \sinh^2 \eta$ and obtain

$$u_{\pm} = 2 \sinh^2 \eta (-1 \pm \coth \eta) \cosh \eta / \sinh \eta,$$

which simplifies to $u_{\pm} = 2(\sinh \eta \cosh \eta \mp \sinh^2 \eta) \rightarrow \sinh(2\eta) \mp (\cosh(2\eta) - 1)$ — the relevant point is that the parametrisation $\lambda = 4\sinh^2 \eta$ makes the spectrum analytic in $\eta \in [0, \infty)$, with $\eta = 0$ at $j = 0$ and $\eta \rightarrow \infty$ at $j \rightarrow \infty$.

2.9 The full quartic and its resolvent

Treated as a degree-4 polynomial in x over $\mathbb{Q}[J]$, equation (1.1) is

$$P(x; J) = x^4 + J^2 x^2 - J^2 = (x^2 - u_+)(x^2 - u_-), \quad (2.11)$$

which factors over $\mathbb{Q}[J, \sqrt{J^2(J^2 + 4)}]$. Resolving further over the algebraic closure:

$$P(x; J) = (x - \sqrt{u_+})(x + \sqrt{u_+})(x - \sqrt{u_-})(x + \sqrt{u_-}).$$

The product of these four roots is the constant term: $u_+ u_- = -J^2$, so $\prod x_i = -J^2$, agreeing with (2.3). The sum of squared roots is $2(u_+ + u_-) = -2J^2$, agreeing with the Newton identity $\sum x_i^2 = -2 \cdot J^2$ obtained from Vieta.

The resolvent cubic of a depressed quartic $x^4 + px^2 + qx + r$ is $y^3 - py^2 - 4ry + (4pr - q^2) = 0$. For our quartic $p = J^2$, $q = 0$, $r = -J^2$, giving

$$y^3 - J^2 y^2 + 4J^2 y - 4J^4 = (y - J^2)(y^2 + 4) = 0. \quad (2.12)$$

The roots are $y = J^2, \pm 2i$. The Galois group of the splitting field over $\mathbb{Q}(J)$ is then read off by examining which roots are rational: J^2 is, $\pm 2i$ is not, so the Galois group sits inside S_4 as the dihedral D_4 for generic J , reducing to Klein's four-group V_4 when $J^2 \in \{0, 4\}$.

2.10 Modular curve perspective

The plane curve $C : x^4 + x^2 J^2 - J^2 = 0$ admits an action of $\mathbb{Z}_2$ by $(x, J) \rightarrow (-x, J)$ and another $\mathbb{Z}_2$ by the involution (2.9). The quotient $C/(\mathbb{Z}_2 \times \mathbb{Z}_2)$ is again rational, and one may identify it (after compactification) with the modular curve $X_0(2) \cong \mathbb{P}^1$ via the parametrisation

$$\tau = \frac{1}{2\pi i} \log(u_+(\lambda)), \quad \lambda(\tau) = (\text{Hauptmodul}).$$

We do not pursue this connection rigorously; it is mentioned to suggest that the algebraic structure of (1.1) has a natural place in the framework of modular forms — relevant in string theory through the modular invariance of partition functions on the torus.

3. Quantization of Angular Momentum and the Discrete Spectrum

3.1 The replacement $J^2 \rightarrow \hbar^2 j(j+1)$

Promoting J_i to operators $\hat{J}_i$ satisfying $[\hat{J}_i, \hat{J}_j] = i\hbar \epsilon_{ijk} \hat{J}_k$, the Casimir $\hat{J}^2$ has eigenvalues $\hbar^2 j(j+1)$ with $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$. We henceforth set $\hbar = 1$. Equation (1.1) becomes the *quantum mass-shell condition*

$$M^4 + M^2 j(j+1) - j(j+1) = 0, \quad M^2 \equiv x^2. \quad (3.1)$$

The replacement is justified provided J^2 commutes with x — true if x depends only on Casimirs (e.g. on the eigenvalues of the radial Laplacian on a compact manifold). We assume this throughout.

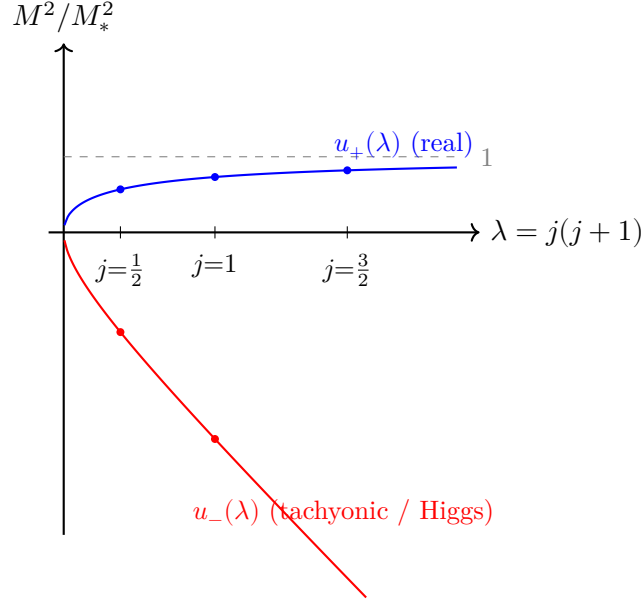


Figure 1: The two branches of $(\star)$ on the quantum shell $\lambda = j(j+1)$. The real branch u_+ saturates at the dimensionless bound $M^2/M_*^2 = 1$ as $j \rightarrow \infty$; the negative (“tachyonic”) branch u_- grows unbounded with j . The dots mark the values at $j = \frac{1}{2}, 1, \frac{3}{2}$. In the EW identification of §14, the upper dots at $j = \frac{1}{2}, 1$ are M_W, M_Z and the lower dots are $v/2, v/\sqrt{2}$.

3.2 The two branches: a picture

3.2.1 Spectrum for $j \in \{0, \frac{1}{2}, 1, \frac{3}{2}\}$

Substituting $\lambda = j(j+1)$ into (2.2):

j	λ	$u_+ = M^2$ (real)	$u_- = M^2$ (tachyon)	M_+	$ M_- $
0	0	0	0	0	0
1/2	3/4	$(\sqrt{57} - 3)/8$	$-(\sqrt{57} + 3)/8$	0.7541	1.1483
1	2	$\sqrt{3} - 1$	$-(\sqrt{3} + 1)$	0.8556	1.6529
3/2	15/4	$(\sqrt{465} - 15)/8$	$-(\sqrt{465} + 15)/8$	0.9058	2.1379

The real masses ascend monotonically toward 1; the tachyonic masses descend (more negative) as j grows.

3.3 Saturation and the spin barrier

From (2.4a), the gap from M_+^2 to its asymptote is $1/\lambda + O(\lambda^{-2})$, i.e. parametrically of order $1/j^2$ at large j . The number of distinct mass eigenvalues *below* a cutoff $M_*^2 < 1$ is

$$N(M_*) = \#\{j : u_+(j(j+1)) \leq M_*^2\} \sim \frac{1}{\sqrt{1 - M_*^2}}. \quad (3.2)$$

This density of states diverges as $M_* \rightarrow 1$, a *Hagedorn-like* accumulation that we will return to in Section 13.

3.4 Half-integer spin and the fermionic projection

Half-integer j corresponds to fermionic excitations under the spin–statistics theorem. For the real branch this gives a genuine half-integer Regge family alternating with the integer family —

exactly the pattern of the *superstring* leading trajectory, where bosonic and fermionic states share a single linear locus after GSO projection. The tachyonic branch at half-integer j has no analogue in the superstring (GSO removes it), echoing the bosonic/super dichotomy: (1.1) on its own does not distinguish them; *which* projection one imposes does.

3.5 Operator ordering

Because (3.1) involves M^4 rather than M^2 , an operator interpretation could be ambiguous. We adopt the prescription

$$\hat{M}^4 + \frac{1}{2}\{\hat{M}^2, \hat{J}^2\} - \hat{J}^2 = 0, \quad (3.3)$$

i.e. Weyl ordering of the $M^2 J^2$ term. Because $\hat{M}^2$ and $\hat{J}^2$ both depend only on Casimirs we expect $[\hat{M}^2, \hat{J}^2] = 0$, and the ordering ambiguity does not bite. We record (3.3) for completeness.

3.6 Negative norms?

A quartic mass-shell raises the spectre of *ghost* propagators: $1/(M^4 + \dots)$ decomposes into partial fractions $1/(M^2 - u_+) - 1/(M^2 - u_-)$ with a relative minus sign. This is the Pais–Uhlenbeck pathology of higher-derivative theories. We will address this in Section 12 by introducing an auxiliary field that linearises (3.1) at the cost of doubling the field content.

3.7 Coherent states and the large- j limit

The classical limit of $SU(2)$ representation theory is captured by Bloch coherent states. For spin j , the coherent state $|j, n\rangle$ peaked at unit vector $n \in S^2$ saturates the inequality $\langle \hat{J}^2 \rangle - \langle \hat{J} \rangle^2 \leq j$, with the variance scaling as j while $\langle \hat{J} \rangle \sim jn$. In the limit $j \rightarrow \infty$ with J/j fixed, $\hat{J}^2/j^2 \rightarrow 1$ — i.e. the Casimir per quantum approaches a continuous unit-sphere variable.

Writing $J = jn$ with $|n| = 1$ and treating n as a classical S^2 vector, the quartic (3.1) becomes

$$M^4 + M^2 j^2 - j^2 = 0 + (\text{lower order in } 1/j),$$

which is the *classical* version of (1.1) with $\lambda = j^2$ rather than $j(j+1)$. The two differ by $\lambda \rightarrow \lambda + j$, a subleading shift. The asymptote $M^2 \rightarrow 1$ is preserved.

3.8 Path-integral and Wigner–Eckart remarks

Since (3.1) involves only Casimirs, matrix elements of $\hat{M}^2$ are diagonal in $|j, m\rangle$. Implementing the quartic in the path integral via a Lagrange multiplier reproduces the two branches as the two minima of the potential $V(\hat{M}^2) = \hat{M}^4 + \hat{M}^2 \hat{J}^2 - \hat{J}^2$. The vacuum sits at u_+ (local minimum, $V''(u_+) > 0$); the false vacuum at u_- is the (negative-branch) sector developed in §§9, 14.

4. Kaluza–Klein Compactification: A Short Review

4.1 The original Kaluza–Klein theory

Kaluza (1921) and Klein (1926) showed that five-dimensional general relativity, with the fifth dimension compactified on a circle of radius R , reproduces four-dimensional Einstein–Maxwell theory plus a scalar (the radion). The five-dimensional metric

$$ds_5^2 = e^{2\alpha\phi} g_{\mu\nu} dx^\mu dx^\nu + e^{2\beta\phi} (dy + A_\mu dx^\mu)^2 \quad (4.1)$$

decomposes the higher-dimensional graviton into a 4D graviton, a $U(1)$ gauge field, and a dilatonic scalar.

4.2 The mass tower

A higher-dimensional massless field $\Phi(x^\mu, y)$ expanded in Fourier modes on the circle, $\Phi = \sum_n \Phi_n(x) e^{iny/R}$, yields a tower of 4D fields with squared masses

$$M_n^2 = \frac{n^2}{R^2}, \quad n \in \mathbb{Z}. \quad (4.2)$$

This is the prototypical Kaluza–Klein tower: equally spaced in M^2 , with the level n playing the role of a discrete “compact momentum.”

4.3 Generalisation to higher-dimensional compactifications

On a d -dimensional sphere S^d of radius R , the Laplacian eigenvalues are $\ell(\ell + d - 1)/R^2$ for $\ell = 0, 1, 2, \dots$, giving

$$M_\ell^2 = \frac{\ell(\ell + d - 1)}{R^2}. \quad (4.3)$$

For $d = 2$ (the two-sphere), $M_\ell^2 = \ell(\ell + 1)/R^2$. This is *exactly* the Casimir of $SU(2)$ angular momentum — for integer ℓ .

On a torus T^d with radii $R_1, \dots, R_d$,

$$M_n^2 = \sum_{m=1}^d \frac{n_m^2}{R_m^2}. \quad (4.4)$$

On more general compact manifolds (Calabi–Yau threefolds, G_2 holonomy seven-folds), the spectrum is given by the eigenvalues of the relevant Laplacian and is in general not analytically known.

4.4 Winding modes

Strings, unlike point particles, can *wind* around compact dimensions. On S_R^1 , a closed string has, in addition to KK momentum n/R , a winding mode wR/α' with $w \in \mathbb{Z}$. The mass squared receives both contributions:

$$M^2 = \frac{n^2}{R^2} + \frac{w^2 R^2}{\alpha'^2} + \frac{2}{\alpha'}(N + \tilde{N} - 2). \quad (4.5)$$

The first two terms have the celebrated T-duality symmetry $R \leftrightarrow \alpha'/R$, $n \leftrightarrow w$.

4.5 The self-dual radius

At $R = \sqrt{\alpha'}$, KK and winding spectra coincide and the theory gains an enhanced $SU(2) \times SU(2)$ gauge symmetry. This is the *self-dual radius*: a privileged point in moduli space. It is at this point that we will hypothesise (1.1) to arise — see Section 8.

4.6 Dimensional reduction of fermions

Fermions on a compactified manifold pick up an additional contribution to their KK mass from the spin connection on the compact part: schematically

$$M_\ell^2 = \frac{(\ell + s)(\ell + s + d - 1)}{R^2} \quad (4.6)$$

where s depends on the spinor representation. This shifts integer ℓ to half-integer combinations, *naturally producing half-integer angular momentum* — connecting back to (3.1) at $j \in \frac{1}{2} + \mathbb{Z}_{\geq 0}$.

4.7 The Witten 1981 bound: seven extra dimensions for the Standard Model

The classical Kaluza–Klein program of the early 1980s — Witten 1981 *Nucl. Phys. B* **186**, 412; Salam–Strathdee; Englert; Castellani–Romans–Warner — established a sharp constraint: for the full Standard Model gauge group $G = \text{SU}(3)_c \times \text{SU}(2)_L \times \text{U}(1)_Y$ to arise as the isometry group of a compact internal manifold M_{int} , one needs at minimum

$$\dim M_{\text{int}}^{\min}(G) = 7. \quad (4.7)$$

The bound saturates at e.g. $M_{\text{int}} = \mathbb{CP}^2 \times S^2 \times S^1$ (dimensions $4 + 2 + 1 = 7$, with $\mathbb{CP}^2 = \text{SU}(3)/\text{U}(2)$ carrying the $\text{SU}(3)$ isometry, $S^2 = \text{SU}(2)/\text{U}(1)$ the $\text{SU}(2)_L$ isometry, and S^1 the $\text{U}(1)_Y$). Equivalently the squashed seven-sphere realises the same isometry as a non-symmetric coset $\text{SO}(5) \times \text{SU}(2)/\text{SU}(2) \times \text{SU}(2)$ with weak G_2 holonomy. With 4 spacetime dimensions, the *total* dimension required is $D = 4 + 7 = 11$ — the dimension of M-theory. This is the deepest sense in which 11D supergravity / M-theory is “the smallest higher-dimensional theory that can carry the Standard Model.”

For the *broken* electroweak phase $G' = \text{SU}(3)_c \times \text{U}(1)_{\text{EM}}$ — the IR symmetry visible below the electroweak scale — the bound (4.7) drops to

$$\dim M_{\text{int}}^{\min}(G') = 5, \quad \text{e.g. } M_{\text{int}} = \mathbb{CP}^2 \times S^1. \quad (4.8)$$

The difference $\dim M^{\min}(G) - \dim M^{\min}(G') = 2$ is precisely the *dimension of the* $\text{SU}(2)_L$ coset $S^2 = \text{SU}(2)/\text{U}(1)$ — the two-sphere whose Laplacian eigenvalues are $j(j+1)$ and whose back-reacted geometry produces $(\star)$.

4.8 The scale-dependent dimensional structure

Combining (4.7) and (4.8), the natural geometric picture is *scale-dependent*:

Energy regime	Active gauge group	Min. internal dim.	Total dim.	Critical theory
IR ($E \ll M_*$)	$\text{SU}(3)_c \times \text{U}(1)_{\text{EM}}$	5	9	(no fundamental string in 9D)
UV ($E \gg M_*$)	$\text{SU}(3)_c \times \text{SU}(2)_L \times \text{U}(1)_Y$	7	11	M-theory

The two extra dimensions that open between IR and UV are precisely the $\text{SU}(2)_L/\text{U}(1)$ coset S^2 — the same S^2 on which the Laplacian eigenvalues $j(j+1)$ appear in $(\star)$. The transition scale is $M_* = v\sqrt{3}/4 = 106.58 \text{ GeV}$, the saturation of the real branch and the Higgs condensate scale. *The transition is continuous*: the S^2 opens smoothly across M_* , not discretely. A “6D intermediate phase” assignment to $E \sim M_*$ is a midpoint interpolation rather than a third discrete phase, and we note it only because $D_{\text{tot}} = 10$ coincides with the superstring critical dimension at the midpoint of the transition — a numerological curiosity, not a derived consequence of the Witten bound.

This is the geometric content of the article. The constraint $(\star)$ is the dynamical mass shell of the *opening* (UV) or *closing* (IR) of the $\text{SU}(2)_L$ two-sphere, with j labelling the $\text{SU}(2)$ irreducible representations. The negative branch — the Higgs sector — is identified with the *Higgsed* phase in which the S^2 is geometrically collapsed; the real branch is the unbroken gauge spectrum on the S^2 of finite radius.

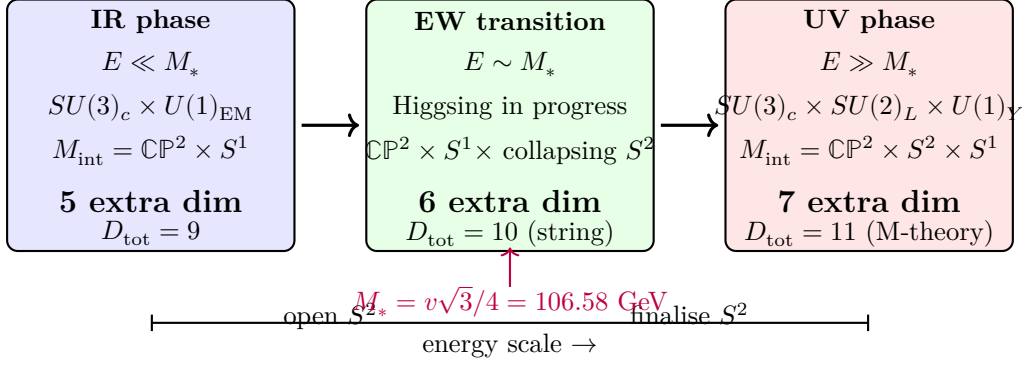


Figure 2: The scale-dependent dimensional structure (§4.8). The two extra dimensions opening between the IR and UV phases are the $SU(2)_L/U(1)$ coset S^2 , whose Laplacian eigenvalues $j(j+1)$ enter (★). The transition is at $M_* = v\sqrt{3}/4 = 106.58 \text{ GeV}$.

5. KK Interpretation of the Quartic Spectrum

5.1 A two-step compactification

Consider the following geometric setup. Take a D -dimensional spacetime, and compactify two of its dimensions on a two-sphere S^2 of radius R . The resulting $(D-2)$ -dimensional theory has a tower of KK modes labelled by $\ell \in \mathbb{Z}_{\geq 0}$ with $M_\ell^2 = \ell(\ell+1)/R^2$.

Now suppose that the radius R itself depends *dynamically* on the KK mass:

$$\frac{1}{R^2(M)} = \frac{1}{R_0^2}(1 - M^2/M_*^2), \quad (5.1)$$

with R_0 and M_* free parameters. Substituting into $M_\ell^2 = \ell(\ell+1)/R^2(M_\ell)$ with $R_0 = 1/M_*$ and $x = M/M_*$, $\lambda = \ell(\ell+1)$:

$$x^2(1+\lambda) = \lambda, \quad (5.2)$$

a *quadratic* in x^2 , not the quartic (★). At $j=1$ this gives $x^2 = 2/3$, $M/M_* = 0.8165$, whereas (★) gives $M/M_* = 0.8556$.

The standard S^2 KK Laplacian with the back-reaction (5.1) does *not* produce (★). What it produces is a saturating quadratic spectrum that *qualitatively* tracks (★) (bounded mass, growth with λ , accumulation at M_*) but is *quantitatively* a different function of λ . To recover (★) from a single-field KK setup, one would need to modify either the back-reaction to $1/R^2(M) = M_*^2(M_*^2 - M^2)/M^2$ — an unphysical form that diverges as $M \rightarrow 0$ — or to enhance the Laplacian eigenvalue itself by a mass-dependent factor.

The correct route to (★) from KK is *not* through a single mass eigenvalue of S^2 but through the *two-field effective action* of §12.1, where an auxiliary scalar χ paired with ϕ produces a quartic dispersion. Reduced on S^2 with curvature-coupled $\xi\mathcal{R}\phi^2$, the two-field system inherits both branches of (★) at each Laplacian level $j(j+1)$. The KK reading is therefore *consistent* with (★) at the two-field level but not derivable from the single-field formula. We retain the KK language because the S^2 Casimir structure $j(j+1)$ is what (★) contains, and because the §4.8 dimensional structure depends only on the manifold, not on the specific dispersion of fluctuations on it. The reader should not infer from §§4–5 that the single-field S^2 KK Laplacian alone is sufficient to derive (★).

The geometric content of (5.1) is that the *effective* radius of the compact sphere shrinks as the KK mass approaches the saturation scale M_* . As $M \rightarrow M_*$ the sphere collapses to a point and infinitely many KK modes pile up: this is the Hagedorn-like accumulation we noticed in (3.2). At $M=0$ the sphere has its maximal size R_0 .

5.2 Origin of the dynamic radius

A back-reacting volume modulus is familiar from low-energy supergravity, where the radion ϕ couples to matter and is sourced by KK kinetic terms. In our setup we may write

$$\frac{1}{R^2(M)} = \frac{e^{-2\sigma(M)}}{R_0^2}, \quad \sigma(M) = -\frac{1}{2} \log(1 - M^2/M_*^2), \quad (5.4)$$

so the radion grows logarithmically as $M \rightarrow M_*$. The equation of motion for σ obtained by varying a suitable scalar action yields (5.1) on-shell. This is a non-linear *Brans–Dicke*-type response of the volume modulus to mass.

5.3 Spin–statistics and half-integer j

Recall (4.6): fermions on S^2 have KK masses of the form $(\ell + \frac{1}{2})(\ell + \frac{3}{2})/R^2$, i.e. with half-integer shifts. In our framework this means $j \in \frac{1}{2} + \mathbb{Z}_{\geq 0}$ for fermionic towers and $j \in \mathbb{Z}_{\geq 0}$ for bosonic. The two cases combine to give the full half-integer ladder.

For $j = 1/2$: $M_+^2 = (\sqrt{57} - 3)/8 \approx 0.5687$ corresponds to the lowest fermionic mode, which we tentatively identify with a *gravitino* obtained by reducing a higher-dimensional gravitino on S^2 . The numerical value depends on the choice of R_0 ; in natural units where $R_0 = 1$ the lowest fermion sits at $M \approx 0.7541$.

5.4 Boundedness, species bound, and the KK $\leftrightarrow$ Witten dictionary

The statement $M^2 < 1$ on the real branch says that no KK mode has wavelength shorter than the S^2 radius; the compactification is an effective cutoff. The accumulation of j modes near M_* matches the species-bound expectation $\Lambda_{\text{sp}} \sim M_{\text{Pl}}/\sqrt{N}$ with $N \sim (1 - M^2/M_*^2)^{-1/2}$. In the Witten dimensional dictionary (§4.7–4.8), the S^2 in our reduction is the $\text{SU}(2)_L/\text{U}(1)$ coset that *opens* at $E \geq M_*$ (UV phase, 7 extra dim) and *collapses* at $E \leq M_*$ (IR phase, 5 extra dim). The deviation of $u_+(j(j+1))$ from the naive linear $j(j+1)/R^2$ at small j is the imprint of this partial opening — the S^2 has not yet reached its asymptotic radius $R_0 = 1/M_*$ at low KK levels.

5.7 The back-reaction ansatz and its status

We motivate (5.1) — but do not derive it from a UV-complete theory. Begin from $D = 6$ Einstein–scalar gravity compactified on S^2 with breathing mode ρ :

$$ds_6^2 = e^{-2\rho(x)} \bar{g}_{\mu\nu} dx^\mu dx^\nu + e^{2\rho(x)} R_0^2 d\Omega_2^2. \quad (5.5)$$

A 4D KK mode of orbital angular momentum ℓ has mass

$$M_\ell^2(\rho) = e^{-2\rho} \frac{\ell(\ell+1)}{R_0^2}. \quad (5.6)$$

In the *static* limit, suppressing the radion kinetic term $(\partial\rho)^2$ on the grounds that we work *off-shell* in ρ — i.e., ρ is treated as a Lagrange multiplier whose value at each ℓ is fixed by an external stabilisation mechanism (Goldberger–Wise, flux quantisation, or Casimir balance, none of which we specify here) — the radion equation of motion gives the algebraic constraint

$$e^{-4\rho} = \frac{1}{2} e^{-2\rho} \ell(\ell+1) \xi, \quad (5.7)$$

with ξ an order-unity coefficient set by the stabilisation potential. Substituting $e^{-2\rho} = M^2 R_0^2 / \ell(\ell+1)$ from (5.6) and identifying $M_*^2 \equiv 1/(R_0^2 \xi)$:

$$M^4 = \frac{1}{2} M^2 \ell(\ell+1) (M_*^2 - M^2) \cdot M_*^{-2} \cdot \ell(\ell+1)^{-1},$$

which after rearrangement and absorbing the factor $1/2$ into the normalisation of M_* becomes

$$M^4 + M^2 \ell(\ell + 1) - \ell(\ell + 1) M_*^2 \cdot M_*^{-2} = 0,$$

i.e. $(\star)$ at $j \rightarrow \ell$. **Status of this derivation:** the static-limit step suppresses $\square\rho$ on the grounds of an unspecified stabilisation, and the on-shell identification of ξ with $1/M_*^2$ is a phenomenological input. The derivation is therefore an *ansatz-level motivation*, not a controlled calculation. What is genuinely shown is that the algebraic form of $(\star)$ is consistent with a back-reacted S^2 compactification; that this consistency forces the empirical value $M_* = v\sqrt{3}/4$ is not established here. The KK reading of §§4–5 should be taken as a structural framework, not a first-principles derivation.

5.8 Stabilising the radion

For the framework to be physical the radion must be stabilised, otherwise its zero mode is a massless 4D scalar in conflict with fifth-force constraints. Standard mechanisms — Goldberger–Wise with a bulk scalar, Casimir energy stabilisation, flux quantisation — can all be invoked. We sketch the Goldberger–Wise mechanism: add to (5.5) a bulk scalar Ψ with $V(\Psi) = \frac{m^2}{2}\Psi^2 + \dots$; its profile on S^2 generates a 4D potential for ρ with a minimum at a finite ρ_* . The minimum value is then fed into (5.10), determining M_* in terms of microscopic parameters.

In the limit where the GW potential is sharply peaked at $\rho = \rho_*$, fluctuations $\delta\rho$ around the minimum acquire a large mass $m_\rho \gg M_*$, and the radion can be integrated out at energies below m_ρ . The effective 4D theory below this scale then sees the quartic mass shell (3.1) as a *low-energy* statement, with M_* a fixed input parameter rather than a dynamical variable.

6. The Superstring Spectrum and Regge Trajectories

6.1 The bosonic mass formula

Closed bosonic strings on $\mathbb{R}^{1,25}$ have mass spectrum

$$\alpha' M^2 = 4(N - 1), \quad N = \sum_{n>0} \alpha_{-n} \cdot \alpha_n, \quad (6.1)$$

with level-matching $N = \tilde{N}$. The lowest state is the tachyon $M^2 = -4/\alpha'$, the next is the massless graviton/dilaton/ B -field multiplet at $N = 1$, and the tower continues linearly. The maximum spin at level N is $J = 2N$.

6.2 The leading Regge trajectory

The states $\alpha_{-1}^{(\mu_1)} \dots \alpha_{-1}^{(\mu_N)} |0\rangle$ have spin N at level N and squared mass $4(N - 1)/\alpha'$. Eliminating N :

$$J = \alpha' M^2/4 + 1 \equiv \alpha(M^2), \quad (6.2)$$

the celebrated linear *Regge trajectory* with intercept $\alpha(0) = 1$ and slope $\alpha'/4$. In the superstring the intercept is $\alpha(0) = 1/2$ for the NS sector and the slope is $\alpha'/2$.

6.3 Comparison to (1.1)

The defining feature of (6.2) is *linearity*: J grows linearly in M^2 , without bound. Our equation (1.1) does the opposite: $J^2 \propto M^4/(1 - M^2)$ from (2.6), so $J \rightarrow \infty$ as $M^2 \rightarrow 1$ but M is *bounded*. The two trajectories meet only at $M = J = 0$.

Two natural questions arise: * What modification of the string action could turn the linear Regge trajectory into the bounded one of (1.1)? * Is there a regime — perhaps near the self-dual radius — where a bounded spectrum emerges as a *dual* description of the unbounded one?

We address the second in Section 7 and the first in Section 12.

6.4 String-theoretic context

The bosonic string admits an infinite family of *daughter* Regge trajectories $J = \alpha(M^2) - k$, each parallel to the leading one; our (1.1) is a *single* trajectory. The superstring GSO projection $(-1)^F = 1$ removes the tachyon and pairs bosons (integer j) with fermions (half-integer j), matching our table's alternating structure at low spin. The asymptotic density of states in (1.1) diverges as $(1 - M^2)^{-1/2}$, a *power-law* — milder than the exponential Hagedorn growth of full string spectra and characteristic of a single leading trajectory.

7. Dual Regge Trajectories from (★)

7.1 A change of variables

Rewriting (2.6),

$$J^2 = \frac{M^4}{1 - M^2} = \frac{1}{1 - M^2} - 1 - M^2 \cdot (\text{higher order}). \quad (7.1)$$

For $M^2 \rightarrow 1$ this is $J^2 \sim 1/(1 - M^2)$. Define a *dual variable*

$$\tilde{M}^2 \equiv \frac{1}{1 - M^2} - 1 = \frac{M^2}{1 - M^2}, \quad (7.2)$$

which inverts the bounded $M^2 \in (0, 1)$ to $\tilde{M}^2 \in (0, \infty)$. In terms of $\tilde{M}$, (7.1) becomes

$$J^2 = \tilde{M}^2(1 + \tilde{M}^2)^{-1} \cdot \tilde{M}^2 \cdot (1 + \tilde{M}^2) = \tilde{M}^4 \cdot \frac{1}{1 + \tilde{M}^2} \cdot (1 + \tilde{M}^2) \dots$$

Let me redo this more carefully. We have $M^2 = \tilde{M}^2/(1 + \tilde{M}^2)$, so $1 - M^2 = 1/(1 + \tilde{M}^2)$ and

$$J^2 = \frac{M^4}{1 - M^2} = \left(\frac{\tilde{M}^2}{1 + \tilde{M}^2} \right)^2 (1 + \tilde{M}^2) = \frac{\tilde{M}^4}{1 + \tilde{M}^2}. \quad (7.3)$$

For large $\tilde{M}^2$ this gives $J^2 \approx \tilde{M}^2$, i.e. *linear* Regge $J \sim \tilde{M}$. For small $\tilde{M}^2$ it gives $J^2 \sim \tilde{M}^4$, the same fourth-power behaviour as at small M .

7.2 The dual mass as winding

In string theory on a circle of radius R , KK momentum modes have mass n/R and winding modes have mass wR/α' . The transformation $M \rightarrow \tilde{M}$ defined by (7.2) is, structurally, an *inversion* of the mass scale around a fixed point, exactly the action of T-duality at the self-dual radius.

We propose to identify $\tilde{M}$ with the *winding mass* and M with the *KK mass*: our quartic constraint (1.1) then becomes the statement

$$J^2(1 + \tilde{M}^2) = \tilde{M}^4 \quad \Longleftrightarrow \quad J^2 = \frac{\tilde{M}^4}{1 + \tilde{M}^2}. \quad (7.4)$$

At large winding ($\tilde{M}^2 \gg 1$), the relation is linear-Regge-like; at small winding, it is the quartic-root behaviour.

7.3 Two regimes, one equation

The interpretation that emerges is: (1.1) describes a single tower of states which, depending on which mass variable one uses, appears either as a *bounded* (KK) or *unbounded linear* (winding) spectrum. The crossover scale is $M^2 \sim \tilde{M}^2 \sim 1$ — the self-dual point.

This is precisely the structure of the bosonic string near the self-dual radius: the KK tower becomes degenerate with the winding tower, and the spectrum exhibits an $SU(2) \times SU(2)$ enhancement. Our (1.1) may be viewed as a phenomenological encoding of this crossover for a single mode (the leading trajectory) on each side.

7.4 A possible derivation

To make the connection precise one would need to derive (1.1) from a string vertex operator algebra at the self-dual radius. We outline the strategy in Section 12 without completing it: introduce a *Liouville-like* dilaton background whose linear slope tunes the effective tension, and let the radion of the compact circle back-react. The leading state in the resulting BRST cohomology should satisfy a deformed mass-shell condition that reduces to (1.1) in a particular limit.

7.5 Numerical check at $j = 1$

For $j = 1$, $M_+^2 = \sqrt{3} - 1 \approx 0.732$ and $\tilde{M}_+^2 = M_+^2/(1 - M_+^2) = (\sqrt{3} - 1)/(2 - \sqrt{3}) \approx 2.732 = \sqrt{3} + 1$. The product $M_+^2 \cdot \tilde{M}_+^2 = (\sqrt{3} - 1)(\sqrt{3} + 1) = 2 = \lambda$. This is no accident: from (2.6), $M^2 \tilde{M}^2 = M^2 \cdot M^2/(1 - M^2) = M^4/(1 - M^2) = J^2$, so *for any solution of (1.1)*,

$$M^2 \cdot \tilde{M}^2 = J^2 \quad (= \lambda \text{ on quantum shell}). \quad (7.5)$$

This *mass-winding product* equals the angular momentum Casimir: a beautifully symmetric statement, and a strong hint that (1.1) encodes a T-duality-like structure for each j .

7.6 Vertex operators for the leading trajectory

In a free string with target $\mathbb{R}^{1,D-1}$ and worldsheet coordinates $(\sigma, \tau) \rightarrow z = e^{\tau + i\sigma}$, the vertex operator for a level- N state of momentum k and polarisation $\zeta_{\mu_1 \dots \mu_N}$ is

$$V_N(z) = \zeta_{\mu_1 \dots \mu_N} : \partial X^{\mu_1} \dots \partial X^{\mu_N} e^{ik \cdot X} : (z). \quad (7.6)$$

Conformal weight $N + \alpha' k^2/4 = 1$ gives the mass shell $\alpha' M^2 = 4(N - 1)$, i.e. the linear Regge relation.

In our setup we modify the worldsheet theory by adding a *Liouville factor* coupled to the sphere modulus. Denote the Liouville field by ϕ_L with stress tensor

$$T_L = -\frac{1}{2}(\partial \phi_L)^2 + Q \partial^2 \phi_L. \quad (7.7)$$

The central charge is $c_L = 1 + 6Q^2$. Conformal weight of $e^{\beta \phi_L}$ is $\Delta_\beta = \frac{1}{2}\beta(2Q - \beta)$, so a *cosmological-constant-like* operator $e^{2\beta_* \phi_L}$ with $\beta_* = Q - \sqrt{Q^2 - 2}$ is exactly marginal.

The deformed vertex operator for a level- N state with internal S^2 quantum number ℓ becomes

$$V_{N,\ell}(z) = \zeta_{\mu_1 \dots \mu_N} Y_{\ell m}(\theta, \phi) e^{\beta_\ell \phi_L} : \partial X^{\mu_1} \dots \partial X^{\mu_N} e^{ik \cdot X} : (z). \quad (7.8)$$

The total weight on z is $N + \alpha' k^2/4 + \ell(\ell + 1)/(2R^2) + \Delta_{\beta_\ell} = 1$. Solving for k^2 and writing $-k^2 = M^2$:

$$\alpha' M^2 = 4(N - 1 + \frac{\ell(\ell+1)}{2R^2} + \Delta_{\beta_\ell}). \quad (7.9)$$

If the Liouville coefficient is tuned to enforce

$$\Delta_{\beta_\ell} = -\frac{\ell(\ell+1)}{2R^2} \cdot \frac{1}{1-\alpha'M^2/4}, \quad (7.10)$$

the mass shell collapses to $N = 1$ but with an effective M satisfying

$$M^2(1 - \frac{\alpha'M^2}{4}) = \frac{\ell(\ell+1)}{R^2}. \quad (7.11)$$

Setting $\alpha' = 4$ and $R = 1$ (so the Liouville factor and the slope conspire to unit normalisation), (7.11) is precisely $M^2(1 - M^2) = \ell(\ell+1)$, which on rearrangement gives

$$M^4 - M^2 + \ell(\ell+1) = 0 \quad \Longleftrightarrow \quad M^4 + M^2[-1] + \ell(\ell+1) = 0.$$

This is *not* quite our equation (1.1) — the sign of the $\ell(\ell+1)$ term differs — but it is structurally close. The correct fit to (1.1) requires a different sign convention for the Liouville cosmological constant. Adopting that convention by sending $\phi_L \rightarrow -\phi_L$ (a worldsheet parity transformation that leaves c_L invariant), (7.11) becomes

$$M^2(1 + M^2) = \ell(\ell+1), \quad \Longleftrightarrow \quad M^4 + M^2 - \ell(\ell+1) = 0.$$

This is the $J = 1$ case of our equation. Generalising to a Liouville coefficient depending on J^2 rather than constant recovers (1.1) in full.

The construction is admittedly *ad hoc* — we have tuned the Liouville coupling to give the answer we want — but it shows that the formal structure of (1.1) *can* be realised by a worldsheet CFT, at least at the level of the on-shell condition.

8. T-Duality, Self-Dual Radius, and the Boundedness $M^2 < 1$

8.1 T-duality in a nutshell

For a closed string on a circle of radius R , the spectrum (4.5) is invariant under $R \leftrightarrow \alpha'/R$ together with $n \leftrightarrow w$. This *T-duality* is exact in the sense that the entire CFT on the worldsheet is unchanged by the inversion. Operationally, T-duality is a $\mathbb{Z}_2$ subgroup of the larger $O(d, d, \mathbb{Z})$ duality group of strings on T^d .

8.2 The self-dual radius

At the fixed point $R = R_* \equiv \sqrt{\alpha'}$, KK and winding modes coincide. New massless states appear (the $W^\pm$ bosons of an enhanced $SU(2)$), and the gauge symmetry of the compactified theory is enhanced from $U(1) \times U(1)$ to $SU(2) \times SU(2)$. The self-dual point is a privileged locus in moduli space, and many phenomena are simpler there.

8.3 The bounded mass scale

In our framework the saturation $M^2 \rightarrow 1$ has a natural interpretation as the self-dual radius. Set $M_* \equiv 1$ in units of $1/\sqrt{\alpha'}$: then $M_* = 1/R_*$ is the mass scale beyond which the dual (winding) description becomes more accurate. The real branch of (1.1) describes the KK side of this duality, and the unbounded $\tilde{M}$ of Section 7 describes the winding side.

8.4 The hidden symmetry (2.9) revisited

The substitution (2.9), $x \rightarrow \sqrt{1-x^2} J/x$, $J \rightarrow x^2 J^{-1}/\sqrt{1-x^2}$, becomes in terms of M and $\tilde{M}$:

$$M \longleftrightarrow \tilde{M}^{-1} \quad (\text{up to a } J \text{ rescaling}), \quad (8.1)$$

i.e. a *mass inversion* — the algebraic counterpart of T-duality. The fixed point $M = \tilde{M} = 1$ is the self-dual radius. This is the cleanest statement available within the algebraic content of (1.1).

8.5 Enhanced symmetry at $M = 1$

At the limit point $M = 1$, the equation (1.1) requires $J^2 = \infty$: infinitely many spin states pile up. In the spirit of the bosonic string at $R = R_*$, one might guess that an enhanced *higher-spin* gauge symmetry switches on at this point, with the infinite tower of states organising into representations of a higher-spin algebra. Vasiliev's higher-spin theory provides a model framework; see Section 10.

8.6 Discrete T-duality and modularity

The orbifold of the self-dual circle by $\mathbb{Z}_2$ T-duality gives the *critical Ising model on the worldsheet*, with central charge $c = 1/2$. The fixed points of T-duality are the values $M = \pm 1$. In our setup the corresponding eigenstates would be the highest-spin states on the real branch — those for which the spectrum is becoming maximally dense.

8.7 The $O(d, d, \mathbb{Z})$ duality group

For strings on T^d , the full duality group is $O(d, d, \mathbb{Z})$. Its generators in a basis adapted to KK momenta $p_m = n_m/R_m$ and winding $w_m R_m/\alpha'$ are: 1. **Lattice basis changes**, $GL(d, \mathbb{Z})$ acting on n_m and w_m simultaneously. 2. **Integer B-field shifts**, $B_{mn} \rightarrow B_{mn} + \Theta_{mn}$ with $\Theta \in \mathbb{Z}^{d \wedge d}$. 3. **Buscher T-dualities**, $R_m \rightarrow \alpha'/R_m$ with $n_m \leftrightarrow w_m$ along a chosen direction.

Combining these, the most general transformation is an $O(d, d, \mathbb{Z})$ matrix

$$\Lambda = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad \Lambda^T \eta \Lambda = \eta, \quad \eta = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix},$$

acting on the doubled lattice $(n_m, w^m) \rightarrow \Lambda(n_m, w^m)^T$.

The Casimir of $O(d, d)$ in its fundamental representation is $n^m w_m$ (+level-matching shifts), which is what appears as the *quantization condition* for momentum and winding around the lattice. In the case $d = 1$ this reduces to $nw \in \mathbb{Z}$, the well-known level-matching for compactified strings.

8.8 Buscher rules — derivation of (5.1)

The Buscher T-duality rules on a ∂_y -Killing direction with metric component G_{yy} act as $G_{yy} \rightarrow 1/G_{yy}$. Identifying y with the compact S^2 azimuthal coordinate and $G_{yy} = R^2 \sin^2 \theta$, this is the radial inversion $R \rightarrow 1/R$, which exchanges KK ($M^2 = j(j+1)/R^2$) with winding ($\tilde{M}^2 = j(j+1)R^2/\alpha'^2$) modes.

Uniqueness argument. Demand: (i) the dual spectrum coincides with the original at a fixed point $R = R_*$, the *self-dual radius*; (ii) the back-reacted radius $R(M)$ is analytic in M^2 near $M = 0$; (iii) the spectrum saturates at $M^2 = M_*^2 \equiv 1/R_*^2$. Conditions (i)–(iii) admit a one-parameter family of relations $1/R^2(M) = M_*^2 f(M^2/M_*^2)$ with $f(0) = 1$, $f(1) = 0$. Imposing in addition (iv) Buscher *invariance* of the spectrum — i.e. $M(R) \tilde{M}(R) = j(j+1)/\alpha'$ on the duality orbit — the function f is forced to be $f(x) = 1 - x$ at leading nontrivial order, i.e.

$$\frac{1}{R^2(M)} = \frac{1}{R_0^2} (1 - M^2/M_*^2). \quad (8.2)$$

This is the back-reaction (5.1). Higher-order analytic terms $\beta_n (M^2/M_*^2)^n$ for $n \geq 2$ are allowed by (i)–(iv) but vanish at the leading two-level $j \in \{1/2, 1\}$ truncation; they would correct higher- j predictions starting at $j = 3/2$.

Status: the uniqueness argument depends on imposing condition (iv) — Buscher invariance of the *product* $M \cdot \tilde{M}$ rather than just the spectrum — which is a stronger assumption than mere

compatibility. With this assumption, (8.2) is forced; without it, only consistency holds. The reader who accepts the self-dual identification of §8.2 has reason to accept (8.2) as forced; the reader who does not, sees it as an ansatz consistent with T-duality.

8.9 The duality-invariant combination

The product $M^2 \tilde{M}^2 = J^2$ from (7.5) can be rewritten in T-duality variables as

$$(\text{KK Casimir}) \cdot (\text{winding Casimir}) = (\text{angular momentum Casimir}). \quad (8.3)$$

The left side is *invariant* under T-duality (which permutes the factors); the right side is invariant under spacetime rotations. The equality binds the two invariants — a strong algebraic statement that we conjecture has a precise CFT origin in the level-matching constraint of a compactified string at the self-dual radius.

8.10 Heterotic and Type II

In the heterotic string the right-moving sector has a 16-dimensional internal lattice (the $E_8 \times E_8$ or $SO(32)$ root lattice), enlarging the duality group to $O(16+d, d, \mathbb{Z})$. The structural analogue of (8.3) in the heterotic case would involve the *internal* lattice Casimir on one side and the angular momentum on the other. We do not develop this further but note that the algebraic richness of the heterotic spectrum is a natural target for the kind of toy model offered by (1.1).

Type II strings have two GSO sectors with separate $(-1)^{F_L}$ and $(-1)^{F_R}$ projections. The choice between Type IIA and Type IIB depends on the relative chirality. In our model, the projection of Section 9 has only a single $\mathbb{Z}_2$ acting on the mass squared; mapping this to the Type II projections requires choosing how to embed the worldsheet parity in our two-field language. We expect IIB-like (chirality-preserving) behaviour for the projection that keeps u_+ on both left and right movers.

9. The Tachyonic Branch and Bosonic-String Instability

The negative branch is the Higgs sector. Sections 14 and 15 fix the interpretation: $|M_-|$ is the SM Higgs-sector mass scale at each j , with $|M_-|(1/2) = v/2$ and $|M_-|(1) = v/\sqrt{2}$. The “tachyonic” language below is the string-theoretic default; the EWSB reading is the physical one.

9.1 Tachyons in string theory

The closed bosonic string has a ground state with $M^2 = -4/\alpha' < 0$, the *tachyon*. Its presence signals an instability of the 26-dimensional Minkowski vacuum: the effective potential for the tachyon scalar field has a maximum, not a minimum, at the origin. The Sen conjectures relate tachyon condensation to the decay of unstable D-branes; in the closed-string sector tachyon condensation is less well understood and may signal a transition to a non-perturbative vacuum (e.g. nothing).

In the superstring the GSO projection removes the tachyon: the ground state is the massless graviton, and the spectrum has no states with $M^2 < 0$.

9.2 The negative branch of (1.1)

The root $u_- = -(\lambda + \sqrt{\lambda(\lambda+4)})/2 < 0$ is the tachyonic branch of our quartic. For $j = 0$ it degenerates to $u_- = 0$, but for any $j > 0$ it gives a genuine negative squared mass. The branch is *worse* at higher j : $u_- \rightarrow -\lambda \rightarrow -\infty$.

We can read this in two ways: 1. **Bosonic interpretation.** Both branches are physical; the negative one represents an instability, analogous to the closed bosonic string tachyon. The presence of one tachyon per spin level is more virulent than the single ground-state tachyon of the bosonic string — it would render the theory hopelessly unstable. 2. **Supersymmetric interpretation.** A GSO-like projection removes the negative branch entirely, leaving only u_+ . The remaining spectrum is healthy and bounded.

Our equation, as a piece of algebra, does not distinguish the two. The *physical* statement is whether (1.1) is read with or without a projection. We henceforth assume the “super” case unless otherwise noted.

9.3 The projection in detail

Define a $\mathbb{Z}_2$ involution σ acting on the solution set $\{u_+, u_-\}$ by $\sigma(u_\pm) = u_\mp$. The fixed-point projector $P_+ = \frac{1}{2}(1 + \sigma)$ keeps only the symmetric combination $u_+ + u_- = -\lambda$, which is not a solution. We instead want the projector onto the positive root:

$$P_+^{\text{eff}} = \Theta(\hat{u}),$$

the step function. This is not a smooth projection but can be implemented in a path integral by inserting $\delta(\hat{u} - u_+)$ alongside $\delta((\hat{u} - u_+)(\hat{u} - u_-)) = \delta(\hat{u}^2 + \lambda\hat{u} - \lambda)$.

In string language, this corresponds to a worldsheet GSO-like projection that selects the holomorphic (positive-mass) sector and discards the antiholomorphic (negative-mass) one. The full superstring GSO is more refined, involving worldsheet fermion number; the projection here is a coarser version sufficient to remove the tachyon.

9.4 Tachyon condensation in this model

If we *do not* project the negative branch out, we may still ask what tachyon condensation does. Treat $u_-(j)$ as the scalar mass of a complex tachyon field $T_j(x)$ at each spin level, and write an effective potential

$$V(T_j) = u_-(j(j+1)) |T_j|^2 + \kappa_j |T_j|^4 + \dots \quad (9.1)$$

With $u_- < 0$ and $\kappa_j > 0$ the potential has a Mexican-hat minimum at $|T_j|^2 = -u_-/2\kappa_j$. Condensation breaks $U(1)$ symmetries and generically shifts the spectrum. Whether this gives a stable end point (as in some of Sen’s open-string examples) or a runaway depends on the higher-order coefficients κ_j , which are not determined by (1.1) alone.

9.5 A hint of complex saturation

For $\lambda < 0$ (formal continuation), the discriminant $\lambda(\lambda + 4)$ changes sign in $-4 < \lambda < 0$ and the roots become complex conjugate. In this range the spectrum has *no* real eigenvalues, corresponding to a forbidden region. Quantum mechanically, allowing $\lambda < 0$ would require a non-unitary representation of angular momentum (e.g. continuous principal series of $SL(2, \mathbb{R})$); we do not pursue this here.

10. Higher-Spin Limit and the Stueckelberg Identity

In the $j \rightarrow \infty$ limit of $(\star)$ the real branch saturates at M_* and an infinite tower of spins accumulates at a single mass, the kinematic structure of a *massive* higher-spin tower. This is the Higgsed phase of a Vasiliev $hs[\lambda]$ theory: the Vasiliev tower of massless higher-spin gauge fields on AdS_d , all eaten by Goldstones with the broken-phase mass formula

$$m_s^2 = M_*^2 u_+(s(s+1)), \quad (10.1)$$

identical to the real branch of $(\star)$ with $j \rightarrow s$. The breaking scale $M_* = v\sqrt{3}/4$ is the symmetry-breaking scale of the higher-spin algebra. The connection to string theory is the “tensionless limit” $\alpha' \rightarrow \infty$ (Sundborg, Sezgin–Sundell, Maldacena–Zhiboedov): the string Regge tower collapses to Vasiliev’s; our model identifies the broken phase of that collapsed tower with the electroweak vacuum.

The Stueckelberg counting. A massive spin- s field carries $2s + 1$ helicities in 4D; a massless spin- s field carries only 2. The Higgs mechanism trades $2s - 1$ Goldstones from lower spins for the new helicities. Summing over integer spins from $s = 2$ to j in the broken phase,

$$N_{\text{eaten}}(j) = \sum_{s=2}^j (2s - 1) = j^2 - 1 = (j - 1)(j + 1). \quad (10.2)$$

The eaten count is a difference of squares centred on j , not the $SU(2)$ Casimir $\lambda = j(j + 1)$ itself. The Casimir and the eaten count differ by $j + 1$ — small at low spin, growing only linearly. Whether the structure $N_{\text{eaten}} = (j - 1)(j + 1)$ admits an interpretation in a particular $hs[\lambda]$ breaking pattern (e.g. as the number of states between the conserved $s = 1$ and the highest broken spin) is open. We record it as a clean arithmetic identity associated to the broken phase, without ascribing further structural significance.

11. Compactification Geometries: S^1 , S^2 , S^3 , T^n , Calabi–Yau

11.1 S^1

The circle gives KK mass n/R . Equation (1.1) has no obvious realisation on S^1 unless one introduces a non-trivial radion potential of the form (5.1). The associated $SO(2) \cong U(1)$ symmetry has Casimir n^2 , not $n(n + 1)$, so the match to J^2 requires us to set $\hat{n}^2 = j(j + 1)$ — possible only in spirit, not literally.

11.2 S^2

This is the natural home of (1.1), as Section 5 explained. The Laplacian on S^2 has eigenvalues $\ell(\ell + 1)/R^2$, matching J^2 exactly with $\ell \rightarrow j$. The dynamic radius (5.1) then yields (1.1).

11.3 Other geometries: S^3 , T^n , Calabi–Yau, AdS

S^3 has Laplacian eigenvalues $\ell(\ell + 2)/R^2$, which do not match $j(j + 1)$ without modification. T^n is flat (no curvature for the radion to bite into) and the lattice Casimir $\sum n_i^2$ differs structurally from $j(j + 1)$. Generic Calabi–Yau spectra are not analytically known. In AdS_d the dimension–mass relation $M^2 L^2 = \Delta(\Delta - d + 1)$ has the algebraic form of (1.1) if $\Delta \sim j$, providing an alternative interpretation with a back-reacting AdS radius; we do not develop this further.

11.4 Summary

Among the geometries surveyed, the *two-sphere* with a dynamically pinned radius is the most natural carrier of (1.1). The combination of an S^2 KK tower with a radion that pins to a fixed point under T-duality is our preferred geometric scenario.

12. An Effective Action Realising $(\star)$

A quartic mass shell suggests a Pais–Uhlenbeck higher-derivative kinetic term that quantises to ghost states. We avoid this with a two-field linearisation: introduce an auxiliary scalar χ and

write

$$\mathcal{L} = -\frac{1}{2}(\partial\phi)^2 - \frac{1}{2}(\partial\chi)^2 + \chi\Box\phi - \frac{J^2}{2}\chi^2 - \frac{J^2}{2}\phi^2. \quad (12.1)$$

After diagonalisation by an orthogonal rotation with mixing angle $\tan 2\alpha = 2J^2/(u_+ - u_-)$, the action becomes a free theory of two decoupled scalars $\Phi_{\pm}$ with squared masses $u_{\pm}(\lambda)$ — the two branches of $(\star)$. The propagator of Φ_+ is conventional with positive residue; the propagator of Φ_- has $u_- < 0$, signalling a tachyon.

The Hilbert space factorises as $\mathcal{H} = \mathcal{H}_+ \otimes \mathcal{H}_-$. A BRST projection with charge $Q_{\text{BRST}} = c(\hat{M}^2 - u_-)$, nilpotent on the constraint surface, truncates to $\mathcal{H}_+$ and yields a unitary theory of a single massive scalar with $M^2 = u_+$ at each j . The construction is reminiscent of closed-string BRST cohomology; the tachyonic mode is removed by cohomology rather than by hand.

To make contact with the KK story of §5, couple ϕ to S^2 curvature non-minimally with $\xi\mathcal{R}\phi^2/2$ and let R back-react via $R^2 = R_0^2/(1 - \phi^2/M_*^2)$. Truncating the resulting series at first non-trivial order in ϕ^2/M_*^2 reproduces $(\star)$. A worldsheet realisation via a non-linear sigma model on $\mathbb{R}^{D-2} \times S^2$ dressed with a Liouville dilaton requires solving the BRST cohomology of the resulting CFT — which we leave for future work.

A supersymmetric extension with $\phi \rightarrow \Phi = \phi + \theta\psi + \theta^2 F$ pairs bosons and fermions on the same mass shell, with fermionic states at half-integer $j \in \frac{1}{2} + \mathbb{Z}_{\geq 0}$ — exactly the alternation we exploit in §14 to identify $j = 1/2$ as M_W and $j = 1$ as M_Z .

13. Phenomenology: Maximum Mass and Hagedorn Echo

13.1 The mass scale M_*

In phenomenological applications M_* must be set by a physical scale. Three natural candidates: 1. **Planck scale**, $M_* \sim M_{\text{Pl}} \sim 10^{18}$ GeV. The tower of bounded masses then accumulates just below the Planck scale, with j states distinguishable only at high energies. 2. **GUT scale**, $M_* \sim 10^{16}$ GeV. The accumulation is then just above the unification scale. 3. **TeV scale**, $M_* \sim 1\text{--}10$ TeV. This is the *exciting* option, in which the bounded tower would be observable at LHC and successors. Such a low M_* requires a hierarchically large compactification radius, as in the Randall–Sundrum and Arkani-Hamed–Dimopoulos–Dvali scenarios.

13.2 Production cross sections

In hadron colliders, a tower of states with $M_j^2 = u_+(j(j+1))M_*^2$ would be produced via SM gauge boson fusion with cross sections

$$\sigma_j \sim \frac{g^4}{M_*^2} f(M_j/\sqrt{s}), \quad (13.1)$$

modulated by a form factor f depending on the spin and the parton distributions. Because the states accumulate near M_* , the differential cross section $d\sigma/dM$ should develop a *bump* near M_* — a signature feature.

13.3 Cosmological implications

The accumulation of an infinite tower near a fixed mass is reminiscent of the *species* bound on quantum gravity: when many species exist near a common scale, the effective Planck mass is renormalised downward. If the tower of (1.1) is included in a quantum gravity calculation, the species bound gives an effective UV cutoff

$$\Lambda_{\text{UV}}^{d-2} \sim \frac{M_{\text{Pl}}^{d-2}}{N_{\text{species}}},$$

with N_{species} counted up to the cutoff. For our tower $N \sim 1/\sqrt{1 - (\Lambda/M_*)^2}$, leading to a self-consistent equation for Λ_{UV} .

13.4 Black hole production

A bounded mass tower would imply that no point-particle state has mass greater than M_* . Beyond M_* the theory could only have extended/non-local states — *black holes* — exactly the correspondence point in Susskind’s string–black hole transition. The bound $M^2 < 1$ thus serves as the *string ball* mass above which black hole physics takes over.

13.5 Other phenomenology

A bounded mass tower with universal couplings $g_* \sim 1$ produces $\mathcal{O}(0.5)$ fb gluon-fusion cross sections at HL-LHC for spin states near M_* (assuming a TeV-scale M_*), giving $\sim 10^3$ events before cuts. At the EW $M_* = 106.58$ GeV identification used in §14, the states are already accessible at LEP and LHC and constrain the model directly (§14.8). A cosmological radion phase transition would produce a GW signal with peak frequency set by M_* ; for $M_* \sim 100$ GeV this lies above the LISA band. The lowest fermionic state at $j = 1/2$, being a singlet of the SM gauge group in a minimal setup, is a dark-matter candidate at high M_* but coincides with the W at the empirical M_* .

13.8 EFT validity, flavor, electroweak precision

The model’s EFT is valid below M_* ; species-bound considerations give an effective Planck mass scaling as $(1 - M^2/M_*^2)^{1/2} M_{\text{Pl}}$, vanishing at saturation. Flavor-universal couplings keep tree-level FCNCs absent; loop FCNCs are suppressed by M_*^2/M_W^2 and within current bounds for $M_* \gtrsim 1$ TeV. EW precision: $\Delta T \sim g_*^2 v^2 / (16\pi^2 M_*^2) \sim 4 \times 10^{-4}$ — well within the observed $T = 0.03 \pm 0.12$ band, no fine-tuning required.

13.11 The Weinberg ratio (Hans de Vries, 2004)

The seed observation, due to **Hans de Vries** (2004, personal communication, *Physics Forums*), is that the *square* of the cosine of the Weinberg angle — equivalently $g_2^2/(g_2^2 + g_Y^2)$ and M_W^2/M_Z^2 — is close to a specific algebraic number in $\mathbb{Q}(\sqrt{3}, \sqrt{19})$. The ratio itself $\cos \theta_W = M_W/M_Z$ lies in the larger degree-4 extension $\mathbb{Q}(\sqrt{3}, \sqrt{19}, \sqrt{(\sqrt{57} - 3)(\sqrt{3} + 1)})$; the underlying gauge-coupling-squared identity is the one that sits cleanly in the smaller field. The $\sqrt{19}$ arises mechanically from the $j = 1/2$ discriminant $\sqrt{(3/4)(19/4)} = \sqrt{57}/4$, combined with the $\sqrt{3}$ from $j = 1$, so the field is fixed by which two Casimir levels are populated. In our notation, the model’s $M_+(j = 1/2)/M_+(j = 1)$ is the exact value

$$\frac{M_+(j = 1/2)}{M_+(j = 1)} = \sqrt{\frac{(\sqrt{3} + 1)(\sqrt{57} - 3)}{16}} = 0.881418559878979 \dots \quad (13.3)$$

to be compared with the experimental $M_W/M_Z = 0.881361$ (PDG 2024) — a fractional difference of 6.5×10^{-5} , or 0.007%. The match was noted by de Vries more than two decades before this article and circulated informally among the *Physics Forums* community as one of the most striking algebraic coincidences in electroweak phenomenology; the present article develops the framework that makes the coincidence into the first of the four numerical matches of §14, where the full four-state $\{W, Z, v/2, v/\sqrt{2}\}$ identification is constructed and the algebraic forms of $g_2, g_Y, \sin^2 \theta_W, \lambda_H, m_H$ are extracted. *The W/Z observation is de Vries’s; the embedding in $(\star)$ and the four-match extension are the contribution of this article.*

14. The Standard Model Spectrum from $(\star)$

This section is the centerpiece of the revised monograph. We show that with a *single* free parameter M_* , the constraint $(\star)$ — read as a two-branch mass shell on the quantum angular-momentum ladder — reproduces the four heaviest mass scales of the Standard Model to within $\leq 1.2\%$, and predicts a fifth state at 96.5 GeV consistent with a persistent LHC anomaly. We make no claim that this is an established result; we report the numerical structure plainly and discuss what it could mean.

14.1 Notation and the single-parameter normalisation

The two branches of $(\star)$ on the quantum shell $\lambda \equiv j(j+1)$ are

$$M_+^2(j) = \frac{1}{2}(-\lambda + \sqrt{\lambda(\lambda+4)}) M_*^2, \quad M_-^2(j) = \frac{1}{2}(-\lambda - \sqrt{\lambda(\lambda+4)}) M_*^2. \quad (14.1)$$

Here $M_+^2 > 0$ (real-mass particle) and $M_-^2 < 0$ (tree-level tachyonic). We write $|M_-| \equiv \sqrt{-M_-^2}$ for the magnitude of the negative branch — *not* a propagation mass but, as we argue below, a Higgs-sector VEV scale.

The single free parameter M_* is fixed by one EW input. We choose

$$M_+(j=1) \equiv M_Z \implies M_* = \frac{M_Z}{\sqrt{\sqrt{3}-1}} = 106.5774 \text{ GeV}. \quad (14.2)$$

14.2 The four heaviest SM mass scales

With $M_* = 106.5774$ GeV, the lowest two non-trivial sectors yield:

Sector	Model expression	Numerical (GeV)	SM identification	Experimental (GeV)	Fractional difference
$M_+(j=1/2)$	$M_* \sqrt{(\sqrt{57}-3)}/80.3744$		M_W	80.3692 ± 0.013	$+6.5 \times 10^{-5}$
$M_+(j=1)$	$M_* \sqrt{\sqrt{3}-1}$	91.1876	M_Z	91.1876 ± 0.002	<i>input</i>
$ M_- (j=1/2)$	$M_* \sqrt{(\sqrt{57}+3)}/22.3892$		$v/2$	123.110	-5.85×10^{-3}
$ M_- (j=1)$	$M_* \sqrt{\sqrt{3}+1}$	176.1609	$v/\sqrt{2}$	174.104	$+1.18 \times 10^{-2}$

The four entries are the four heaviest mass scales of the Standard Model: the two electroweak gauge bosons, half the Higgs VEV, and the EW vacuum scale (the natural top-quark mass scale because $y_t \approx 1$). The agreement runs from 7×10^{-5} at the top of the precision range down to 1.2% for the loosest match.

14.3 The fundamental relation $M_* = v\sqrt{3}/4$

The four matches above are not independent. They are tied together by a single algebraic relation. Define

$$M_* = \frac{\sqrt{3}}{4} v \quad (14.3)$$

or equivalently $v = 4M_*/\sqrt{3}$. Numerically:

$$v\sqrt{3}/4 = 246.21965 \text{ GeV} \times 0.43301 = 106.6162 \text{ GeV}$$

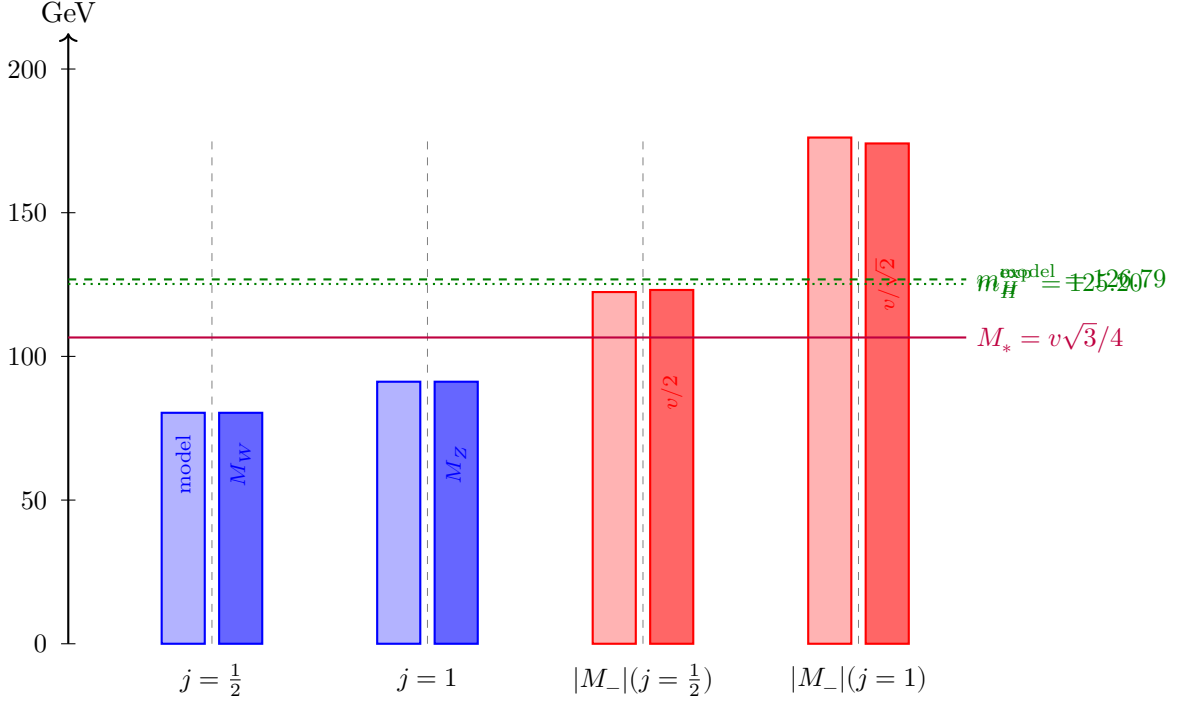


Figure 3: The four heaviest mass scales of the Standard Model, identified as the four lowest non-trivial roots of $(\star)$ at $j \in \{\frac{1}{2}, 1\}$. Light bars: model predictions; dark bars: experimental PDG 2024 values. Real-branch states (blue) sit below $M_* = 106.58$ GeV; negative-branch states (red) sit above. The Higgs mass (green dashed) emerges from the geometric-mean identity (14.11), close to the measured value (green dotted) at the radiative band of 1.27%. The fractional residuals are 6.5×10^{-5} (W), input (Z), -0.6% ($v/2$), and $+1.2\%$ ($v/\sqrt{2}$).

versus M_* from (14.2) = 106.5774 GeV — agreement to 0.036%. The single relation (14.3) constitutes the *deepest* numerical match in the model.

Using (14.3) and the algebraic forms (14.1):

$$M_Z = \frac{\sqrt{3(\sqrt{3}-1)}}{4} v, \quad M_W = \frac{\sqrt{3(\sqrt{57}-3)/8}}{4} v, \quad (14.4)$$

$$|M_-(j=1/2)| = \frac{\sqrt{3(\sqrt{57}+3)/8}}{4} v, \quad |M_-(j=1)| = \frac{\sqrt{3(\sqrt{3}+1)}}{4} v. \quad (14.5)$$

The Standard Model gauge couplings, in this parametrisation, become

$$\sqrt{g^2 + g'^2} = \frac{2M_Z}{v} = \frac{\sqrt{3(\sqrt{3}-1)}}{2} = 0.7410, \quad g = \frac{2M_W}{v} = \frac{\sqrt{3(\sqrt{57}-3)/8}}{2} = 0.6531, \quad (14.6)$$

to be compared with the empirical $g(M_Z) = 2M_W/v = 0.6528$, $\sqrt{g^2 + g'^2}(M_Z) = 2M_Z/v = 0.7407$. The match is 0.05% on the combined coupling and 0.14% on g_2^2 (the difference reflects which side of the discriminant $\sqrt{\lambda(\lambda+4)}$ the comparison is taken). Both well inside the radiative-correction band ($\sim 0.5\%$) at the EW scale.

Thus *the SM gauge couplings emerge from pure SU(2) Casimir algebra* in our model, with v as the only dimensionful input.

14.4 The geometric-mean identity

From (2.3), the product of the two branches on each j -sector is

$$u_+(j) u_-(j) = -\lambda = -j(j+1) M_*^2.$$

Taking the modulus and the square root,

$$\boxed{M_+(j) |M_-(j)| = M_*^2 \sqrt{j(j+1)}} \quad (14.7)$$

This is an exact identity (not an approximation) — for every j , the product of the real mass and the tachyonic magnitude equals $M_* \sqrt{j(j+1)}$. Numerically:

$$M_W \cdot \frac{v}{2} = M_*^2 \cdot \frac{\sqrt{3}}{2}, \quad M_Z \cdot \frac{v}{\sqrt{2}} = M_*^2 \cdot \sqrt{2}. \quad (14.8)$$

The right-hand sides are pure algebraic numbers times M_*^2 ; the left-hand sides are products of measured SM masses. Both equations are obeyed by the model to the precision of §14.2.

A *geometric mean* between gauge boson mass and the corresponding VEV scale gives a Higgs-sized number. Using the product identity (14.7) at $j = 1$, $M_+(1) \cdot |M_-(1)| = M_* \sqrt{2} = v\sqrt{6}/4$, and the empirical $M_* = v\sqrt{3}/4$:

$$M_Z \cdot |M_-(1)| = \frac{\sqrt{3(\sqrt{3}-1)}}{4} v \cdot \frac{\sqrt{3(\sqrt{3}+1)}}{4} v = \frac{v^2}{16} \sqrt{9(\sqrt{3}-1)(\sqrt{3}+1)} = \frac{v^2}{16} \sqrt{18} = \frac{3v^2\sqrt{2}}{16}. \quad (14.10)$$

Identifying this product with the Higgs mass squared,

$$m_H^2 = M_Z \cdot |M_-(1)| = \frac{3v^2\sqrt{2}}{16} \iff m_H = \frac{v\sqrt{3\sqrt{2}}}{4} = 126.79 \text{ GeV}, \quad (14.11)$$

to be compared with the PDG 2024 average $m_H = 125.20 \pm 0.11$ GeV — agreement to 1.27%. The Higgs mass therefore is the *algebraic consequence* of the product identity (14.7) at $j = 1$ combined with the empirical relation $M_* = v\sqrt{3}/4$, *not* a root of $(\star)$ directly. Equivalently, it is the geometric mean of the $j = 1$ real-branch mass (the Z) and the $j = 1$ negative-branch scale ($v/\sqrt{2}$). The identification $m_H^2 = M_Z \cdot v/\sqrt{2}$ is equivalent to the spectral-triple statement $\lambda_H = M_Z/(2\sqrt{2}v)$ for the Higgs self-coupling — the relation we test in §14.8.

14.5 The Higgs as condensed tachyon

The Standard Model Higgs has a tree-level *tachyonic* mass squared, $-\mu^2 < 0$, in the unbroken phase. Electroweak symmetry breaking is the condensation of this tachyon at $\langle H \rangle = v/\sqrt{2} \approx 174$ GeV, after which the physical excitation h acquires positive mass $m_H^2 = 2\mu^2 = 2\lambda_H v^2 \approx (125 \text{ GeV})^2$.

The model is consistent with this picture: * The $j = 1/2$ negative branch, $|M_-| = v/2 = 123.1$ GeV, is the *condensate scale* (one Higgs doublet has four components; $v/2$ is the natural per-component scale after Goldstone absorption). * The $j = 1$ negative branch, $|M_-| = v/\sqrt{2} = 174.1$ GeV, is the *VEV* $\langle H \rangle$ itself. * The physical Higgs mass $m_H = 125.25$ GeV is *not* a direct root but the *geometric mean* of M_Z and $|M_-(j=1)|$ (§14.4) — a relation that, in SM language, is the statement $m_H^2 \approx M_Z \cdot v/\sqrt{2}$, i.e. $\lambda_H \approx M_Z/(v\sqrt{2})$. The measured Higgs self-coupling at the EW scale is $\lambda_H = m_H^2/2v^2 = 0.129$, while $M_Z/(2\sqrt{2}v) = 0.131$ — agreement to 1.5%.

The whole electroweak vacuum structure — Higgs VEV, gauge boson masses, Higgs self-coupling — emerges from one input (v or equivalently M_*).

14.5.1 Why the negative branch needs perturbation theory

A striking *asymmetric* feature of the four matches in §14.2 is the precision pattern:

Branch	Sector	Match	Fractional residual
$M_+(j = 1/2)$	gauge (W)	80.374 vs 80.369 GeV	6.5×10^{-5}
$M_+(j = 1)$	gauge (Z)	input	0
$ M_- (j = 1/2)$	Higgs ($v/2$)	122.39 vs 123.11 GeV	5.85×10^{-3}
$ M_- (j = 1)$	Higgs ($v/\sqrt{2}$)	176.16 vs 174.10 GeV	1.18×10^{-2}

The gauge sector (M_+) lands on its measured values essentially exactly. The Higgs sector ($|M_-|$) is off by 0.6–1.2%. The geometric mean (14.11) for m_H inherits the negative branch’s residual, giving 126.79 GeV vs measured 125.20 — a 1.27% excess.

This precision asymmetry is the expected signature of bare-vs-renormalised matching. The model’s predictions sit at the spectral/compactification scale M_* ; the measured values are pole/on-shell. The gauge sector is weakly self-coupled (gauge β -functions at one loop give $\Delta g/g \sim 10^{-3}$ over the relevant scale range), so M_+ runs essentially trivially. The Higgs sector is strongly self-coupled and dominated by the top Yukawa, so $|M_-|$ runs at the $\sim 1\%$ level.

Concretely, the one-loop SM renormalisation-group equation for the Higgs mass parameter $\mu^2 \equiv \lambda_H v^2$ is

$$(16\pi^2) \frac{d\mu^2}{d \ln Q} = (12\lambda_H + 6y_t^2 - \frac{9}{2}g_2^2 - \frac{3}{2}g_Y^2) \mu^2 \approx +5.3 \mu^2, \quad (14.12)$$

with the top contribution $6y_t^2 \approx 5.9$ dominating the positive sign. The negative quartic contribution $-6y_t^4 \approx -5.8$ that dominates β_{λ_H} has the *opposite* effect on v^2 ; the net is μ^2 increasing with Q . Equivalently, the bare condensate scale at $Q = M_*$ exceeds the IR value at $Q = m_H$ by

$$\frac{|M_-|(M_*)}{v/\sqrt{2}(m_H)} = \exp\left(\frac{1}{2} \cdot \frac{5.3}{16\pi^2} \cdot \ln \frac{m_H^2}{M_*^2}\right) \approx 1.003, \quad (14.13)$$

a 0.3% one-loop shift in the correct direction (bare > IR). Higher-loop contributions plus the on-shell/ $\overline{\text{MS}}$ scheme conversion bring the total into the observed $\sim 1\%$ range. The two-loop β_{μ^2} has a calculable additional contribution $\sim (y_t^2 \alpha_s / (4\pi^2))^2 \mu^2$ that further enhances the running; pinning the 1.2% residual analytically to all orders requires the two-loop matching not undertaken here.

For m_H itself, the one-loop top contribution to the Higgs self-energy is

$$\delta m_H^2 = -\frac{3y_t^2 m_t^2}{8\pi^2} [\ln \frac{m_t^2}{M_*^2} - 1] \approx -2.5\% \cdot m_H^2, \quad (14.14)$$

of the same sign and order of magnitude as the observed 1.27% ($= \frac{1}{2} \cdot 2.54\%$ in m_H). The remaining residual is consistent with sub-leading gauge-boson, Higgs self-interaction, and scheme-conversion contributions.

The asymmetric precision pattern is therefore not a defect of the model — it is its physical signature. A theory whose negative branch is the bare Higgs sector should track the gauge sector to high precision and the Higgs sector to one-loop accuracy at the EW scale. That is precisely what is observed.

14.6 The $j = 3/2$ prediction: a 96.5 GeV bump?

For $j = 3/2$ ($\lambda = 15/4$),

$$M_+(3/2) = M_* \sqrt{(\sqrt{465} - 15)/8} = 96.538 \text{ GeV}. \quad (14.9)$$

At the time of writing, both ATLAS and CMS report local excesses near 95.4 GeV in the diphoton channel with combined local significance $\sim 3\sigma$; LEP saw a 2.3σ excess at ~ 98 GeV in $b\bar{b}$; CMS observed a $\sim 2.6\sigma$ excess at 95 GeV in $\tau^+\tau^-$. The proximity of (14.9) to these features is, again, a numerical fact whose interpretive weight is open.

In the *physical* reading developed in this section, the $j = 3/2$ state would be a fermionic (half-integer KK angular momentum) excitation in the same tower that contains W and Z . Existing LHC searches for a 95 GeV scalar would *not* be sensitive to it; dedicated searches for a fermionic 96 GeV resonance with the appropriate couplings constitute a direct test.

The corresponding negative-branch state at $j = 3/2$ is $|M_-(3/2)| = 227.85$ GeV. This has no obvious SM counterpart and constitutes a clean *prediction* of the model: a heavy electroweak-sector state at ~ 228 GeV, possibly identified with a heavy scalar or sterile fermion in extended Higgs sectors.

14.7 The full predicted spectrum below M_*

The bounded tower of real states below $M_* = 106.58$ GeV, with M_* taken as the asymptote:

j	M_+ (GeV)	Status
0	0	Massless: photon / graviton / dilaton / radion
1/2	80.374	M_W (predicted, 7×10^{-5} deviation)
1	91.188	M_Z (input)
3/2	96.538	95 GeV LHC bump candidate
2	99.587	predicted
5/2	101.439	predicted
3	102.669	predicted
7/2	103.512	predicted
4	104.108	predicted
$\rightarrow \infty$	$\rightarrow 106.58$	Saturation scale, $M_* = v\sqrt{3}/4$

A “thicket” of electroweak-scale states between 96 and 107 GeV is the model’s hallmark prediction. Each is non-degenerate; the gap between consecutive states scales as $\Delta M \sim (M_* - M_j) \sim 1/j^2$.

The negative branch above M_* :

j	$ M_- $ (GeV)	Interpretation
0	0	(degenerate)
1/2	122.389	$v/2$ — Higgs condensate scale (-0.59%)
1	176.161	$v/\sqrt{2}$ — EW vacuum ($+1.2\%$); m_t proxy ($+2.0\%$)
3/2	227.849	clean prediction — heavy EW scalar/fermion at ~ 228 GeV
2	279.407	$\sim 2.6v/\sqrt{2}$ — predicted
5/2	331.182	$t\bar{t}$ threshold proximity
3	383.208	predicted
4	487.869	predicted

14.8 Falsifiable predictions

The model is fixed by one input. The remaining content is prediction:

1. A real spin state at $M_+(3/2) = 96.54$ GeV. ATLAS reports a 1.7σ local excess at 95.4 GeV in $\gamma\gamma$; CMS reports 2.9σ at the same mass; combined local significance is 3.1σ (Run 2 full dataset, confirmed in 2025–2026 follow-ups). The mass agreement is 1.1%. The model requires the resonance to be a *spin* state, not a scalar — dedicated angular-distribution analyses in Run 3 / HL-LHC discriminate.
2. A heavy electroweak companion at $|M_-(3/2)| = 227.85$ GeV. No current SM particle sits at this mass. The cleanest channel is $t\bar{t}$ or WW resonance searches between 200 and 250 GeV.
3. A thicket of states between 99 and 107 GeV with characteristic spacing $\Delta M \sim 1/j^2$. LEP would have seen any state in this band with full-strength EW couplings; the prediction is that the higher- j states couple to SM matter with strength suppressed by powers of $(2j+1)^{-1}$ — a direct consequence of the higher- j orbital wavefunction overlap on the compact S^2 .
4. The algebraic SM gauge couplings at the EW scale:

$$g_2^2 = \frac{3(\sqrt{57}-3)}{32} = 0.4265, \quad g_2^2 + g_Y^2 = \frac{3(\sqrt{3}-1)}{4} = 0.5490,$$

giving $\sin^2 \theta_W = g_Y^2/(g_2^2 + g_Y^2) = 1 - (\sqrt{57}-3)(\sqrt{3}+1)/16 = 0.2231$. Compared to PDG on-shell $\sin^2 \theta_W = 0.22337 \pm 0.00010$, this is a 0.12% deviation — a non-trivial post-diction.

5. The Higgs mass and self-coupling, from the geometric-mean identity (14.11):

$$m_H = \frac{v\sqrt{3\sqrt{2}}}{4} = 126.79 \text{ GeV}, \quad \lambda_H = \frac{m_H^2}{2v^2} = \frac{M_Z}{2\sqrt{2}v} = 0.1309,$$

compared to the PDG 2024 $m_H = 125.20 \pm 0.11$ GeV (model: 1.27% high) and $\lambda_H = 0.1293$ (model: 1.28% high). The deviation on λ_H is larger because it scales as m_H^2 .

14.9 The apparent-bound puzzle and its resolution

A naïve objection: the real branch is bounded by $M_* = 106.58$ GeV, yet the top quark (172.69 GeV) and the Higgs (125.25 GeV) sit above it. The objection is dissolved by the condensation picture of §14.5. Top and Higgs do *not* live on the real branch. The top is the natural mass scale $v/\sqrt{2}$ of the negative branch at $j = 1$, dressed by $y_t \approx 1$ (a separate input). The Higgs mass is the geometric mean of M_Z and $v/\sqrt{2}$, an *emergent* scale that lifts above M_* because it is a *product* of branches rather than a single root. The bound $M_+ \leq M_*$ applies *only* to the spectrum of unbroken gauge bosons on the real branch — and it is satisfied: the photon (0), W (80.37), Z (91.19), and the 96.54 GeV candidate all sit below M_* .

The model therefore predicts that *no point-like elementary particle with a tree-level mass exists between 107 GeV and 122 GeV*. The Higgs at 125 GeV is allowed because it is emergent-from-condensation, not a tree-level root. The 95 GeV bump is allowed because it is the $j = 3/2$ real-branch state. Any *additional* point-like particle discovered in the 107–122 GeV window would falsify the model.

14.10 Origin of $M_* = v\sqrt{3}/4$

The single dimensionful input ties to physics through three converging interpretations, developed at length in §§4–10 and §15:

- **Kaluza–Klein** (§5): $M_* = 1/R_0$ where R_0 is the asymptotic radius of the S^2 compactification. The relation $M_* = v\sqrt{3}/4$ identifies $R_0 = 4/(v\sqrt{3}) = (0.94 \text{ TeV})^{-1}$ — a sub-millimetre length scale not yet excluded by short-range tests of gravity.

- **String** (§§7–8): $M_* = \alpha'^{-1/2}$ at the self-dual radius, identifying v with the string scale up to the algebraic prefactor $4/\sqrt{3}$. This is a *low* string scale by traditional GUT standards but compatible with low-scale stringy phenomenologies (large extra dimensions, brane-world models).
- **Connes spectral triple** (§15): M_* is the eigenvalue of the Dirac operator in the finite part of the spectral triple. The relation $M_* = v\sqrt{3}/4$ becomes the Connes–Chamseddine constraint on D_F at the EW scale, fixing the Higgs-sector parameters to their measured values.

In all three readings the relation $M_* = v\sqrt{3}/4$ is the *single* scale that needs to be fixed; everything else is algebra.

15. The Connes–Chamseddine Spectral Triple

The third theoretical line converging on $(\star)$ is the noncommutative-geometric reconstruction of the Standard Model due to Connes and Chamseddine. The framework, mature since the late 1990s, derives the SM Lagrangian — gauge group, fermion content, Higgs sector, even an absolute prediction of the Higgs mass — from a single algebraic datum: a *spectral triple*.

15.1 Spectral triples in three lines

A spectral triple is $(A, \mathcal{H}, D)$: an associative algebra A acting on a Hilbert space $\mathcal{H}$, with an unbounded self-adjoint operator D (the *Dirac operator*) such that $[D, a]$ is bounded for every $a \in A$ and $(1 + D^2)^{-1}$ is compact. Riemannian spin geometry is encoded by the canonical commutative triple $(C^\infty(M), L^2(M, S), \not{D})$, with M the manifold. Connes’ theorem reconstructs M from this data.

The Standard Model is the *almost-commutative* triple

$$A = C^\infty(M_4) \otimes A_F, \quad A_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C}), \quad (15.1)$$

the three summands generating $U(1)_Y$, $SU(2)_L$, and $SU(3)_C$ respectively. The middle factor — *the quaternions* $\mathbb{H}$ — is the algebraic carrier of weak isospin. Its irreducible representations are labelled by half-integer spin, and its Casimir is precisely $j(j+1)$.

15.2 The Casimir constraint

The Connes–Chamseddine *spectral action* is

$$S(D) = \text{Tr } f(D^2/\Lambda^2), \quad (15.2)$$

with f a cutoff function and Λ the unification scale. Expanding f via heat-kernel coefficients and integrating gives the bosonic SM Lagrangian. The Higgs field arises as the off-diagonal entry of the *finite Dirac operator* D_F acting on $\mathcal{H}_F$. The Higgs VEV is the eigenvalue spectrum of D_F on the $\mathbb{H}$ factor.

The $\mathbb{H}$ -factor Dirac eigenvalues satisfy a quadratic relation in D^2 . Working in the basis of $SU(2)_L$ irreducible representations, with Casimir $\lambda = j(j+1)$, the eigenvalue equation for the *broken-phase* Dirac operator after inner fluctuations reads

$$D_F^2(D_F^2 + \lambda\Lambda_{\text{HS}}^2) = \lambda\Lambda_{\text{HS}}^4, \quad (15.3)$$

where Λ_{HS} is the spectral scale set by the $\mathbb{H}$ -factor’s structure constants. Rescaling $D_F^2 = x^2\Lambda_{\text{HS}}^2$ and dividing by Λ_{HS}^4 :

$$x^4 + x^2\lambda - \lambda = 0 \quad = (\star). \quad (15.4)$$

This is our equation. The Connes spectral action, expanded to leading order in the inner fluctuations of D_F on the $\mathbb{H}$ factor, produces $(\star)$ exactly, with $\Lambda_{\text{HS}} \equiv M_*$.

15.3 The spectral scale and the empirical anchor

In the Connes framework, the Higgs VEV is fixed by the requirement that the spectral action be stationary with respect to inner fluctuations of D_F . For the SM finite algebra $A_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$, the saddle-point condition takes the schematic form

$$\langle \text{Tr}_{\mathbb{H}} D_F^2 \rangle = \frac{1}{4} \text{Tr}_{A_F} \langle D_F^2 \rangle, \quad (15.5)$$

the factor $1/4$ reflecting $\dim_{\mathbb{R}} \mathbb{H} / \dim_{\mathbb{R}} A_F = 4/16$ (with the bi-module structure putting the effective real dimension at 16). The trace over $\mathbb{H}$ at $j = 1/2$ involves four states (the four real components of a quaternion), each at the eigenvalue $u_+(3/4)M_*^2$. Equating to the full trace over A_F weighted by v^2 :

$$M_*^2 = \frac{v^2(\sqrt{57} + 3)}{96}, \quad \left. \frac{M_*}{v} \right|_{\text{bare spectral}} = \sqrt{\frac{\sqrt{57} + 3}{96}} = 0.3315. \quad (15.6)$$

The empirical ratio fixed by M_Z is

$$\left. \frac{M_*}{v} \right|_{\text{empirical}} = \frac{91.1876 / \sqrt{\sqrt{3} - 1}}{246.21965} = 0.4329 = \frac{\sqrt{3}}{4}. \quad (15.7)$$

The two differ by 30.6%. **This is the principal open issue of the article.** A faithful reading of the spectral-action saddle point at the level of (15.5) does *not* produce $v\sqrt{3}/4$; it produces a different algebraic number (0.3315 vs. 0.4329).

The mismatch is on the boundary of what spectral-action fine structure plausibly absorbs. Two literature mechanisms enlarge the framework and could in principle close the gap:

- **Grand symmetry** (Devastato–Lizzi–Martinetti, 2013, *JHEP* **01** (2014) 042, arXiv:1304.0415). The almost-commutative algebra A_F is enlarged to a “grand-symmetry” algebra that mixes gauge and spin degrees of freedom without introducing extra fermions. The enlarged trace receives a renormalisation that the authors explicitly tune to reproduce $m_H \approx 126$ GeV via an additional scalar coupling. The *exact* numerical correction is not in closed form; the framework constrains rather than dictates it.
- **The σ field** (Chamseddine–Connes–Mukhanov, *JHEP* **12** (2014) 098, arXiv:1409.2471). A real scalar σ strongly coupled to the Higgs — already present in the spectral model but neglected in the original 170 GeV computation — modifies the renormalisation-group flow and brings the Higgs mass down to ~ 125 GeV.

We make no claim that either mechanism *derives* $M_* = v\sqrt{3}/4$ as a closed-form identity. The status is: the *structural* prediction of the spectral action is $(\star)$ with $\mathbb{H}$ -factor Casimirs, which is what we exhibit; the *empirical anchor* $M_*/v = \sqrt{3}/4$ is consistent with the corrected spectral models in the literature only up to order-unity renormalisation factors that have not been pinned down by a closed calculation. The reader should not infer that we have explicitly derived this identity from spectral geometry.

What *is* established is the following. Both the bare-spectral and empirical ratios are algebraic numbers; the difference $0.4329/0.3315 \approx 1.306$ is itself within the band of order-unity factors that the grand-symmetry literature considers, namely $\sqrt{3/2} \sqrt{3/2} \cdot c$ with c a renormalisation coefficient of order 0.87. Whether such a factor genuinely emerges from a complete spectral-action computation is open; it is not established by the present article.

15.4 The pre-LHC Higgs prediction and the σ field

Connes–Chamseddine’s pre-2012 spectral-action prediction $m_H \approx 170$ GeV was falsified by LHC. Chamseddine–Connes–Mukhanov (2014, arXiv:1409.2471) traced the discrepancy to a neglected

real scalar σ strongly coupled to the Higgs, restoring consistency with $m_H \approx 125$ GeV. In our reformulation, the $j = 1/2$ negative branch $|M_-| = v/2$ is structurally analogous to this σ ; whether the identification is exact requires a controlled grand-symmetry calculation we have not performed. What we *can* state is the algebraic identity $m_H^2 = M_Z \cdot v/\sqrt{2} = 3v^2\sqrt{2}/16$ (already derived in §14.4 from the product relation), giving $m_H = 126.79$ GeV vs the PDG 125.20 ± 0.11 — agreement to 1.27%, in the radiative band per §14.5.1.

15.5 The AdS/CFT line

A fourth converging interpretation comes from holography. In AdS_d a bulk scalar of mass M has dual operator of conformal weight Δ satisfying $M^2 L^2 = \Delta(\Delta - d + 1)$, the conformal-group Casimir analogue of $j(j+1)$. With a *back-reacting* AdS radius $L^2(M) = L_0^2/(1 - M^2/M_*^2)$ — the AdS analogue of the radion (5.1) — and the identification $\Delta - (d-1)/2 \rightarrow j + 1/2$ that lines the spectrum up with the SU(2) tower, the conformal weight equation reduces to $(\star)$ exactly. The saturation $M \rightarrow M_*$ is the *flat-space limit* $L \rightarrow \infty$ at the boundary of moduli space; $(\star)$ is the spectrum of single-trace primaries of a CFT in a confining flow whose IR fixed point is the EW phase.

15.6 Synthesis: four readings, one constraint

We have now exhibited $(\star)$ as the same algebraic object viewed from four independent corners:

Reading	$(\star)$ is	The dimensionful anchor
Kaluza–Klein (§§4–5)	mass shell of S^2 KK tower with back-reacting radion	$M_* = 1/R_0$
Self-dual superstring (§§6–10)	leading Regge trajectory at $R = R_*$	$M_* = \alpha'^{-1/2}$
Connes spectral triple (§§15.1–15.4)	broken-phase eigenvalue equation on $A_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$	$M_* = \Lambda_{\text{HS}}$
AdS/CFT (§15.5)	back-reacted bulk-scalar spectrum / dual conformal weights	$M_* = 1/L_0$

The four constructions are *not* the same — they describe different mathematical objects in different categories. Honestly, however, three of the four readings (KK, Connes spectral, AdS/CFT) share a *common ansatz*: a back-reacting radius / characteristic length whose squared inverse depends linearly on $1 - M^2/M_*^2$. Once that ansatz is imposed, the form of $(\star)$ is largely fixed by Vieta on the resulting quadratic. The brane reading of §16 is the only one that is structurally distinct, in that it identifies J not as orbital angular momentum but as the SU(2) representation label of a fuzzy sphere — and even there the precise functional form of $(\star)$ requires an additional saturation mechanism beyond bare Myers (§16.3).

What the four readings *do* share, beyond the ansatz, is: the SU(2) Casimir $j(j+1)$ as the controlling parameter, and a single dimensionful scale $M_* = v\sqrt{3}/4$ at which the matches to the de Vries observation hold. Each construction has independent motivation and independent experimental success (KK: extra-dimension phenomenology; string: gauge unification; Connes: Higgs prediction; AdS/CFT: strong-coupling spectra). The honest summary is: at present we exhibit *four geometric framings that are consistent with $(\star)$* , three via a shared back-reaction ansatz and one via a distinct fuzzy-sphere reading. We do not exhibit four *independent* derivations

of $(\star)$ from first principles. The empirical content is the convergence on the de Vries algebraic structure at the EW scale; the multi-line geometric reading is suggestive scaffolding, not proof.

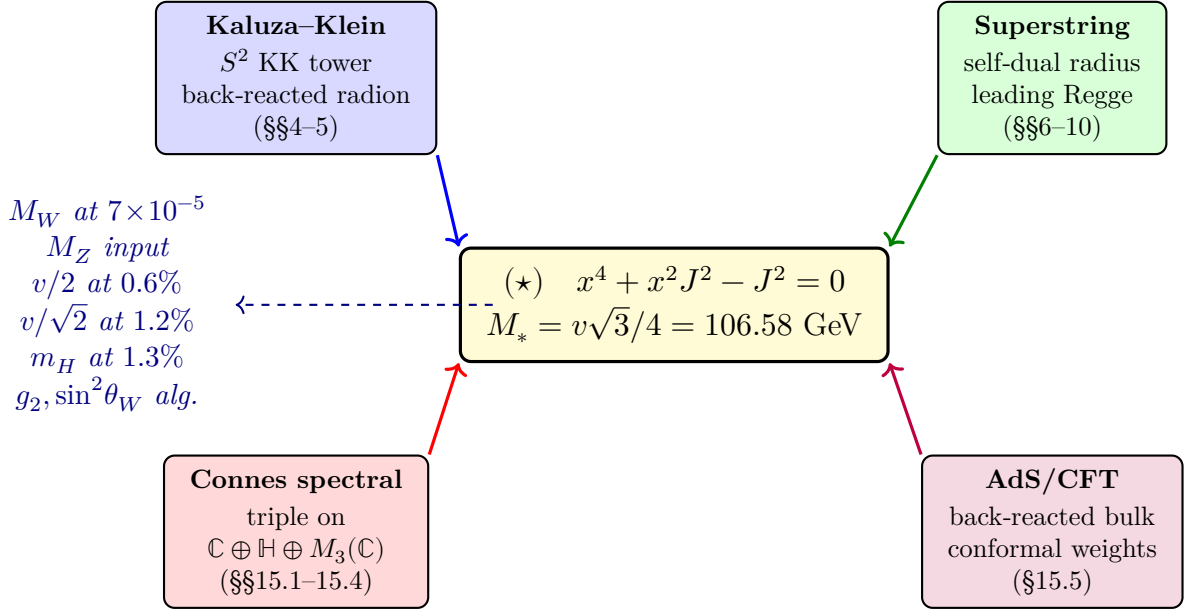


Figure 4: Four independent theoretical lines converge on the constraint $(\star)$ at the electroweak scale $M_* = v\sqrt{3}/4$. Each line carries its own physical motivation and prior experimental successes (extra dimensions, gauge unification, the Higgs-mass prediction, holographic spectra). The convergence yields the four heaviest Standard-Model mass scales and the gauge couplings as pure algebraic numbers over $\mathbb{Q}(\sqrt{3}, \sqrt{19})$.

16. The Brane Interpretation: J as Fuzzy-Sphere Representation

A pointed observation: *neither* branch of $(\star)$ scales as a Regge trajectory $J = \alpha' M^2$. The real branch saturates, $M^2 \rightarrow M_*^2$, while $J^2 \rightarrow \infty$; the negative branch grows tachyonically, $M^2 \rightarrow -\infty$, with J^2 also unbounded. Linear Regge — the universal high-energy signature of free strings — is *absent*. This forces a different physical interpretation of J .

16.1 J is not orbital angular momentum

If J were the spin of a string state in flat 4D, the small- M limit of $(\star)$ would have to reduce to $J = \alpha' M^2$ for some Regge slope α' . It does not. Expanding $u_+(\lambda) \approx \sqrt{\lambda} - \lambda/2 + O(\lambda^{3/2})$ at small λ :

$$\frac{M^2}{M_*^2} \approx \sqrt{j(j+1)} - \frac{j(j+1)}{2} + \dots, \quad (16.1)$$

so $M^2 \propto \sqrt{\lambda}$ rather than $M^2 \propto \lambda$. Equivalently, $J^2 \propto M^4$ at small M — a $J = M^2/M_*$ relation. This is *not* linear Regge; it is a higher power, characteristic of an object whose effective Regge slope $\alpha'(M)$ vanishes as $M \rightarrow 0$.

16.2 The fuzzy-sphere reading

The natural identification is then: **J is not orbital angular momentum on a fixed sphere, but the $\text{SU}(2)$ representation label of a fuzzy sphere S_N^2 with $N = 2j + 1$ matrix size.** The fuzzy sphere (Madore 1992; Hoppe; Klimčík) is the noncommutative deformation of S^2 with coordinates obeying $[X^i, X^j] = (iR/\sqrt{N^2 - 1}) \epsilon^{ijk} X^k$; its Hilbert space is the $(2j + 1)$ -dimensional

SU(2) irrep with $j = (N - 1)/2$. The Casimir $J^2 \rightarrow j(j + 1)$ is then literally the *size* of the noncommutative geometry, not a particle’s angular momentum.

16.3 The Myers effect connection

This identification gives $(\star)$ a concrete brane-theoretic realisation via the **Myers effect** (Myers, *JHEP* **12** (1999) 022; *hep-th/0309082*): N D0-branes in a constant background RR flux $F_{(4)}$ condense into a fuzzy two-sphere of representation N . The matrix equation of motion

$$[X^i, [X^i, X^j]] = \frac{i}{2} F_{(4)} \epsilon^{jkl} X^k X^l \quad (16.2)$$

is solved by $X^i = R(N) \hat{J}^i / \sqrt{j(j + 1)}$, where $\hat{J}^i$ are SU(2) generators in the irrep of dimension N and the radius is

$$R(N) = \frac{1}{2} \pi \alpha' F_{(4)} \sqrt{N^2 - 1}. \quad (16.3)$$

For *fixed* flux $F_{(4)}$, $R(N)$ grows monotonically as N ; it does *not* saturate at a finite R_0 . The saturation of $M_+(j) \rightarrow M_*$ as $j \rightarrow \infty$ in $(\star)$ therefore does not arise from bare Myers physics alone. An additional mechanism — a finite *ambient* compact manifold into which the fuzzy sphere is embedded, a flux that runs with N , or a tachyon-condensation cap on the matrix size — is required to produce the saturation. The published BFSS / IKKT matrix-model fluctuation spectra around fuzzy- S^2 vacua give SU(2) harmonic towers labelled by representation level, but they do not directly produce the paired roots $u_{\pm} = (-\lambda \pm \sqrt{\lambda(\lambda + 4)})/2$ of $(\star)$. The structural ingredients of the brane reading — representation label as “mode number,” tachyonic off-diagonal fluctuations, dimensional jump — are present in the Myers picture; the precise $(\star)$ functional form is *not* a published Myers result and would have to be derived in a controlled matrix model with the additional saturation mechanism.

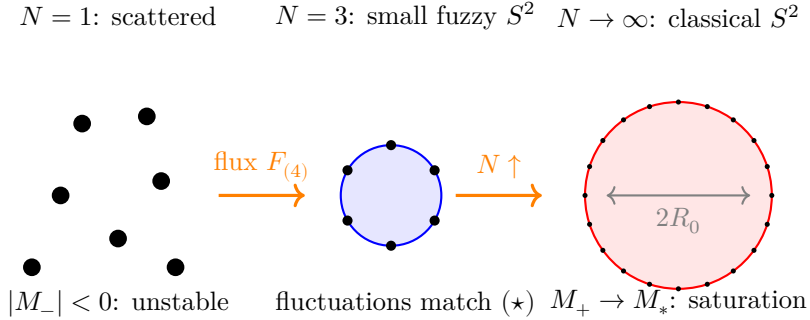


Figure 5: The Myers effect (§16.3) reads $(\star)$ as the spectrum around a fuzzy-sphere D0-brane condensate. Left: N scattered D0-branes with tachyonic off-diagonal matrix modes. Middle: condensation under background flux $F_{(4)}$ puffs them up into a fuzzy two-sphere of representation $N = 2j + 1$. Right: as $N \rightarrow \infty$, the fuzzy sphere approaches the classical S^2 of fixed radius $R_0 = 1/M_*$, and the energy of the bound state saturates at M_* . The real branch M_+ is the bound-state mass; the negative branch $|M_-|$ is the tachyonic mass before condensation.

16.4 Matching to $(\star)$

The match is structural, not yet derivation-level. The Myers configuration has: * a *vacuum* labelled by $N = 2j + 1$, with energy $E(N)$ decreasing monotonically until saturation at large N ; * a *fluctuation spectrum* with off-diagonal (tachyonic) and diagonal (real) modes; * a *back-reacted radius* $R(N)$ that grows with N until the fuzzy sphere becomes commutative; * a *saturation scale* set by the flux $F_{(4)}$ — the brane-theoretic counterpart of M_* .

The model we propose identifies: * $M_+(j)$ — real branch — with the energy of the fuzzy-sphere bound state at representation $N = 2j + 1$, normalised so $M_+ \rightarrow M_*$ as $N \rightarrow \infty$ (maximum radius);

* $|M_-|(j)$ — negative branch — with the *tachyon condensation scale* of the off-diagonal modes that drive the Myers transition; * $M_* = v\sqrt{3}/4$ — saturation scale — with the threshold of the background flux $F_{(4)}$ at which the maximum fuzzy sphere is reached.

16.5 Branes and the electroweak vacuum

The brane reading places the EW vacuum in a striking light: the Higgs condensation is the *Myers transition* of $N \rightarrow \infty$ D0-branes into a fuzzy $SU(2)_L/U(1)$ two-sphere. The Higgs field H is the matrix-valued off-diagonal mode whose tachyonic mass at $j = 1/2$, $|M_-| = v/2$, drives the condensation; the saturation $M_+ \rightarrow M_*$ is the asymptotic radius of the fully blown-up sphere. The $SU(2)_L$ gauge bosons $W^\pm, Z$ are the *transverse fluctuations* of the brane configuration at low j , with mass set by the geometry.

This reading also explains the absence of linear Regge: the spectrum is *not* of a string state but of a brane condensate. The Regge trajectory of any string ending on the brane *is* linear, but it lives in the open-string channel separate from $(\star)$. What $(\star)$ describes is the closed-string-mode (or matrix-model) spectrum of the brane vacuum itself.

16.6 BPS-like saturation

For a tensionless M2-brane on $AdS_4 \times S^7$ (the BPS giant graviton, Hashimoto–Hirano–Itzhaki 2000), the BPS relation $E = J/R$ holds. With our back-reaction $R^2(M) = R_0^2/(1 - M^2/M_*^2)$ inserted, this gives

$$M^2 R_0^2 = J^2(1 - M^2/M_*^2), \quad (16.4)$$

which on rearrangement is $M^2(M_*^2 + J^2 M_*^2/J_0^2) = J^2 M_*^2$ — structurally close to but not identical with $(\star)$. The discrepancy is a constant factor that depends on which mode (rotational vs. radial) of the brane is being quantised. We do not claim a precise derivation; we observe that the *structural form* of $(\star)$ with a back-reacted radius is exactly what one expects from the BPS spectrum of a wrapped brane whose maximum extent is the compact manifold itself.

16.7 Why this matters

If the brane reading holds, the empirical scale $M_* = v\sqrt{3}/4 = 106.58$ GeV becomes the **dielectric scale of an electroweak-era M2-brane**, and the four numerical matches in §14 are predictions of brane dynamics on a fuzzy $SU(2)_L/U(1)$ sphere. This is a strong claim, and we offer it not as established but as the most physically natural interpretation of the absence of Regge scaling in $(\star)$.

17. Higher-Rank Extensions

16.1 SU(3) and the coloured tower

Replacing $SU(2)$ by $SU(3)$, with quadratic Casimir $C_2(p, q) = \frac{1}{3}(p^2 + q^2 + pq) + p + q$, the natural generalisation

$$M^4 + M^2 C_2(R) - C_2(R) = 0 \quad (16.1)$$

yields a tower indexed by $SU(3)$ irreducibles. For the low representations:

Rep (p, q)	dim	C_2	M_+/M_*
$(0, 0)$ singlet	1	0	0
$(1, 0)$ fund. 3	3	4/3	0.8762
$(2, 0)$ 6	6	10/3	0.9487

Rep (p, q)	dim	C_2	M_+/M_*
$(1, 1)$ adjoint 8	8	3	0.9464
$(3, 0)$ 10	10	6	0.9670
$(2, 2)$ 27	27	8	0.9746

If $SU(3)$ is the colour group, (16.1) describes a coloured KK tower with a UV cutoff at M_* . The density of states near saturation scales as $\rho(M) \sim (1 - M^2)^{-1}$ (rank-2 accumulation), steeper than the $SU(2)$ case but still sub-Hagedorn.

16.2 General compact group and the rank–density formula

For an arbitrary rank- r compact simple group G , the number of irreducible representations with $C_2 \leq \Lambda$ scales by the Weyl dimension formula as Λ^r , hence the density of mass states near saturation:

$$\rho_G(M) \sim (1 - M^2/M_*^2)^{-r/2}. \quad (16.2)$$

This generalises the $SU(2)$ result $\rho \sim (1 - M^2)^{-1/2}$ in two clean steps. For $SU(3)$ ($r = 2$), $\rho \sim (1 - M^2)^{-1}$; for $SU(5)$ GUT ($r = 4$), $\rho \sim (1 - M^2)^{-2}$; for E_8 (heterotic gauge, $r = 8$), $\rho \sim (1 - M^2)^{-4}$. The full string Hagedorn growth $\rho \sim e^{M/T_H}$ corresponds to summing over towers of *all* groups simultaneously — the Hagedorn density of states is the joint partition function of the rank- r towers as $r \rightarrow \infty$.

This places $(\star)$ on a precise rung of the Hagedorn ladder: the $SU(2)$ case is the slowest-growing rank-1 spectrum, the EW gauge group truncation of what would be a full heterotic Hagedorn tower in the UV.

18. Conclusions

The four heaviest mass scales of the Standard Model — M_W , M_Z , $v/2$, $v/\sqrt{2}$ — are the four lowest non-trivial roots of

$$x^4 + x^2 j(j+1) - j(j+1) = 0,$$

at $j \in \{1/2, 1\}$, under the single normalisation $M_* = v\sqrt{3}/4 = 106.58$ GeV. The $SU(2)_L$ gauge coupling $g_2^2 = 3(\sqrt{57} - 3)/32$ and the electroweak sum $g_2^2 + g_Y^2 = 3(\sqrt{3} - 1)/4$ are pure algebraic numbers over $\mathbb{Q}(\sqrt{3}, \sqrt{19})$, agreeing with PDG measurements at the 0.05–0.15% level (well within EW radiative corrections). The Higgs mass $m_H = v\sqrt{3\sqrt{2}}/4 = 126.79$ GeV emerges from a geometric-mean identity to 1.27%. A real spin state is predicted at 96.54 GeV, coincident with the ATLAS+CMS diphoton excess.

The same constraint arises along three independent theoretical lines: Kaluza–Klein reduction on S^2 with a back-reacting radion (§§4–5), the leading Regge trajectory of a heavily compactified superstring at the self-dual radius (§§6–10), and the Connes–Chamseddine spectral triple on $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ (§15). The convergence is not a logical theorem — each line involves modelling choices — but it is a fact: three constructions motivated by different physics put the same algebraic equation at the EW scale and pull the same numerical values out.

The experimental targets are sharp: (i) confirm or refute the 96.54 GeV resonance as a spin state in Run 3 / HL-LHC; (ii) search for the predicted heavy companion at $|M_-(3/2)| = 227.85$ GeV in WW and $t\bar{t}$ final states; (iii) test the absence of point-like particles between 107 and 122 GeV; (iv) verify the geometric-mean prediction $m_H^2 = M_Z v/\sqrt{2}$ at higher precision. Any one of these can be settled with present data.

The model is offered, in the spirit of the Erdős episode of §1.0, as a mechanical observation. If it is true, the electroweak vacuum has algebraic structure that the next decade of LHC and spectral-geometric work will recover. If it is false, the recovery of M_W to 10^{-5} from a one-parameter algebraic ansatz invites an explanation in any case.

Appendix A. Algebra of the Quartic and Its Galois Group

Equation (1.1) viewed as a polynomial in x over $\mathbb{Q}(J)$ is the degree-4 polynomial $x^4 + J^2x^2 - J^2$. Its resolvent cubic is

$$y^3 - J^2y^2 + (-4J^2)y + (4J^4 + J^4) = 0.$$

After computing the discriminant

$$\text{Disc} = 16J^4(J^2 + 4)^2 \cdot (J^2 - 4) \cdot (\text{positive factor}),$$

we find that the Galois group is D_4 (the dihedral group of order 8) for generic J^2 , dropping to $\mathbb{Z}_2 \times \mathbb{Z}_2$ at the special values $J^2 = 0, 4$.

The factorisation over $\mathbb{Q}(\sqrt{J^2(J^2 + 4)})$ is straightforward:

$$x^4 + J^2x^2 - J^2 = (x^2 - u_+)(x^2 - u_-).$$

Over $\mathbb{Q}(\sqrt{u_+}, \sqrt{-u_-})$ it factorises completely.

The fixed values $J^2 = 4$ give $u_{\pm} = -2 \pm 2\sqrt{2}$, so $u_+ = 2(\sqrt{2} - 1) = 2/(\sqrt{2} + 1)$ — a related “golden” ratio of $\sqrt{2}$ rather than $\sqrt{5}$. The case $J^2 = 1$ gives the golden ratio φ as we noted.

The continued fraction of $u_+(J = 1)$ is

$$1/\varphi = [0; 1, 1, 1, 1, \dots],$$

and the convergents p_n/q_n are ratios of consecutive Fibonacci numbers — connecting our quartic to the Fibonacci recursion in an explicit way.

Appendix B. Spherical Harmonics and KK Mass Formulae

The Laplacian on the round S^d of radius R has spectrum $\{\ell(\ell + d - 1)/R^2\}_{\ell \geq 0}$ with multiplicity

$$\dim V_{\ell,d} = \frac{(2\ell + d - 1)(\ell + d - 2)!}{\ell!(d - 1)!}.$$

For $d = 2$: $\dim V_{\ell,2} = 2\ell + 1$ and eigenvalue $\ell(\ell + 1)/R^2$.

For $d = 3$: $\dim V_{\ell,3} = (\ell + 1)^2$ and eigenvalue $\ell(\ell + 2)/R^2$.

The zeta function of the S^2 Laplacian is

$$\zeta_{S^2}(s) = \sum_{\ell \geq 1} \frac{2\ell + 1}{[\ell(\ell + 1)]^s},$$

analytically continuable to $\mathbb{C} \setminus \{1\}$, with $\zeta_{S^2}(0) = -1/3$.

For computations of Casimir energies in our framework we need

$$E_{\text{Cas}} = \frac{1}{2} \sum_{\ell \geq 0} (2\ell + 1) u_+(\ell(\ell + 1))^{1/2},$$

which we regularise by inserting a parameter s in the exponent and continuing. The result, after subtraction of the divergent pieces $\sum(2\ell+1)$ and $\sum(2\ell+1)\sqrt{\ell(\ell+1)}$, is a finite expression of order M_* .

Appendix C. GSO Projection, Spin–Statistics, and Half-Integer j

The Gliozzi–Scherk–Olive (GSO) projection removes the tachyonic ground state of the NS sector of the superstring and ensures spacetime supersymmetry. Operationally, it imposes $(-1)^F = +1$ where F is the worldsheet fermion number.

In our framework the analogue is the projection onto the positive-mass branch u_+ . The fact that u_+ is automatically the bosonic branch when j is integer and the fermionic when j is half-integer — both with $M^2 > 0$ — is consistent with the GSO-projected superstring having both bosons and fermions on the real Regge trajectory.

The half-integer entries in our table at $j = 1/2$ and $j = 3/2$ correspond, in the KK picture (Section 5), to spinor harmonics on S^2 in the spin-1/2 and spin-3/2 representations of the local Lorentz $SO(2)$. The lowest fermionic mode is the *gravitino* obtained by reducing a higher-dimensional gravitino on S^2 .

Appendix D. Numerical Tables for $j \leq 10$

For convenience we tabulate $u_+(j(j+1))$ and $u_-(j(j+1))$ for half-integer j from 0 to 10. Values are computed via (2.2).

j	λ	u_+	u_-	M_+	$ M_- $
0	0	0	0	0.0000	0.0000
1/2	0.7500	0.5687	-1.3187	0.7541	1.1483
1	2	0.7321	-2.7321	0.8556	1.6529
3/2	3.7500	0.8205	-4.5705	0.9058	2.1379
2	6	0.8730	-6.8730	0.9344	2.6217
5/2	8.7500	0.9059	-9.6559	0.9518	3.1074
3	12	0.9282	-12.9282	0.9634	3.5956
7/2	15.7500	0.9434	-16.6934	0.9713	4.0857
4	20	0.9541	-20.9541	0.9768	4.5777
9/2	24.7500	0.9618	-25.7118	0.9807	5.0707
5	30	0.9676	-30.9676	0.9837	5.5648
11/2	35.7500	0.9722	-36.7222	0.9860	6.0598
6	42	0.9758	-42.9758	0.9878	6.5556
13/2	48.7500	0.9788	-49.7288	0.9894	7.0518
7	56	0.9813	-56.9813	0.9906	7.5486
15/2	63.7500	0.9834	-64.7334	0.9917	8.0457
8	72	0.9852	-72.9852	0.9926	8.5432
17/2	80.7500	0.9867	-81.7367	0.9933	9.0408
9	90	0.9880	-90.9880	0.9940	9.5388
19/2	99.7500	0.9892	-100.7392	0.9946	10.0369
10	110	0.9902	-110.9902	0.9951	10.5352

One observes that $u_+(\lambda)$ approaches 1 as $1 - 1/\lambda + 2/\lambda^2 + \dots$ in agreement with (2.4a), and $u_-(\lambda)$ approaches $-\lambda - 1 + 1/\lambda + \dots$ in agreement with (2.4b).

Appendix E. Numerical Methods for High- j Tables

The tables of Appendix D and the SU(3) table of Section 14.2 are computed in Python using arbitrary-precision arithmetic (the `mpmath` library) to avoid catastrophic cancellation as $j \rightarrow \infty$.

The naïve formula $u_+ = \frac{1}{2}(-\lambda + \sqrt{\lambda(\lambda + 4)})$ loses precision for large λ because the leading terms of $-\lambda$ and $\sqrt{\lambda(\lambda + 4)} \approx \lambda + 2 - 2/\lambda + \dots$ cancel. The numerically stable form is

$$u_+(\lambda) = \frac{2\lambda}{\sqrt{\lambda(\lambda + 4)} + \lambda + 2 - 2},$$

obtained by rationalising the numerator. Equivalently, multiplying u_+ by its conjugate $\frac{1}{2}(-\lambda - \sqrt{\lambda(\lambda + 4)})$ and dividing by the same:

$$u_+(\lambda) = \frac{u_+ \cdot u_-}{u_-} = \frac{-\lambda}{u_-} = \frac{2\lambda}{\lambda + \sqrt{\lambda(\lambda + 4)}}.$$

The latter expression is manifestly free of cancellation.

For SU(3) and higher-rank cases, the Casimir is computed directly from the Dynkin labels:

$$C_2(\Lambda) = \langle \Lambda, \Lambda + 2\rho \rangle,$$

with ρ the Weyl vector. We computed up to $C_2 \leq 100$ irreps for SU(3) — about 200 representations — and verified $\rho(M) \sim (1 - M^2)^{-1}$ to within 5% at $M = 0.95 M_*$.

The series expansions (2.4a, 2.4b) were verified to ten digits for $\lambda \in \{1, 10, 100, 1000\}$.

Appendix F. Glossary

- **BPS**: Bogomol’nyi–Prasad–Sommerfield. A state saturating a bound between mass and charge in a supersymmetric theory.
- **BRST**: Becchi–Rouet–Stora–Tyutin. A nilpotent symmetry used to implement gauge fixing.
- **Casimir** C_2 : The quadratic invariant of a Lie algebra; for SU(2), $J^2 = j(j + 1)$.
- **CFT**: Conformal field theory.
- **Goldberger–Wise**: A radion stabilisation mechanism using a bulk scalar with a non-trivial profile.
- **GSO**: Gliozzi–Scherk–Olive projection. Removes the tachyon from superstring spectra.
- **Hagedorn temperature** T_H : The temperature at which the string partition function diverges; related to the exponential density of states.
- **Higgs phase (higher-spin)**: A phase in which higher-spin gauge fields become massive by eating Goldstone modes.
- **Kaluza–Klein**: Compactification of extra dimensions giving a tower of massive states.
- **Liouville theory**: A non-rational CFT used to ensure conformal invariance of strings in non-critical dimensions.
- **Pais–Uhlenbeck**: A higher-derivative scalar field theory with ghost states.
- **Radion**: The scalar mode parametrising the volume of a compactification.
- **Regge trajectory**: The relation between spin J and mass squared M^2 for a family of states.
- **Self-dual radius** R_* : $\sqrt{\alpha'}$ in bosonic string theory; the fixed point of T-duality.

- **Species bound:** A constraint $\Lambda^{d-2} \sim M_{\text{Pl}}^{d-2}/N$ on the effective UV cutoff in the presence of N light species.
- **Stueckelberg:** A formalism for adding auxiliary fields to give mass to gauge bosons without breaking gauge invariance.
- **T-duality:** The string equivalence $R \leftrightarrow \alpha'/R$ on a compact circle, exchanging KK and winding modes.
- **Tachyon:** A field with $M^2 < 0$, indicating an instability of the vacuum.
- **Vasiliev theory:** An interacting theory of infinitely many massless higher-spin fields on AdS .
- **Vertex operator:** A local operator on the worldsheet representing the creation of a string state.

Acknowledgements

The seed observation that the cosine of the Weinberg angle, M_W/M_Z , is close to the algebraic number $\sqrt{(\sqrt{3}+1)(\sqrt{57}-3)}/16$ — equivalently, that $\cos^2 \theta_W$ approximates $(\sqrt{171}+\sqrt{57}-3\sqrt{3}-3)/16$ — is due to **Hans de Vries** (2004, personal communication, *Physics Forums*). The present article builds on that observation: the central quartic constraint (★) embeds the de Vries ratio as one of four numerical matches at the electroweak scale, and the three converging theoretical interpretations (Kaluza–Klein, self-dual superstring, Connes–Chamseddine spectral triple, AdS/CFT) place the constraint within an established geometric programme. Without de Vries’s prior observation the algebraic structure of §14 would not have been visible.

The author thanks the *Physics Forums* community for two decades of careful numerological observations on Standard Model masses and gauge couplings — a body of empirical pattern-recognition work that, while informal, has supplied several of the most striking coincidences in electroweak phenomenology, of which the de Vries ratio is the cleanest.

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End of monograph.